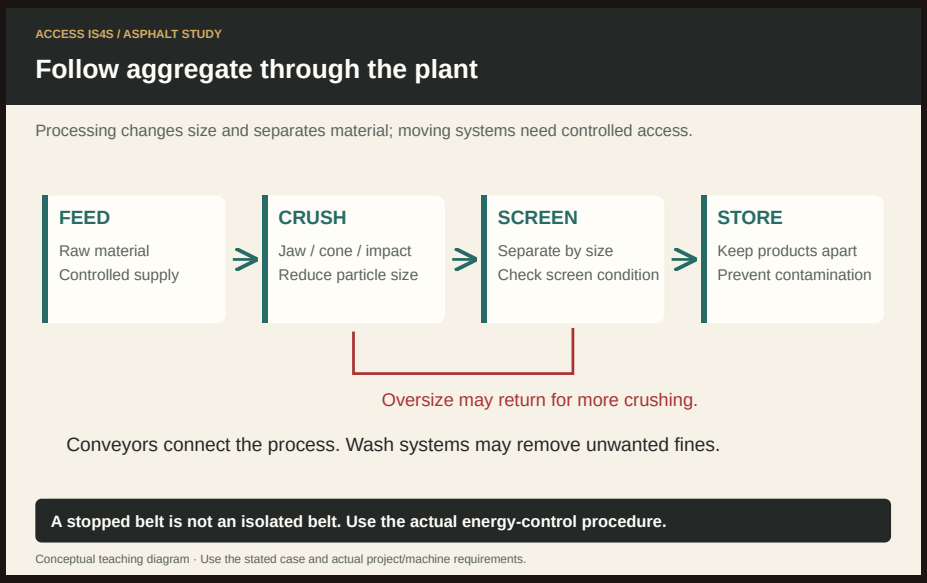
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Materials, plants and quality evidence

Volume 3 · C1, C2, C3, C4, D1, D2, D3, E1, E2, E3, E4

Classroom development edition · 30 October 2026



ACCESS IS4S. (2026). Follow aggregate through the plant [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

In this volume

1. How to use the volume and key terms
2. Crushing process and paving aggregate production
3. Types and specifications of aggregates
4. Asphalt plant operation
5. Asphalt products and uses
6. Environmentally friendly properties
7. Use of storage areas
8. Asphalt plant and placement emissions
9. Determine asphalt mix design
10. Confirm that samples match mix design
11. Determine haul truck samples
12. Determine compaction levels
13. Trace a material and its sample

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for materials, plants and quality evidence. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline C1

Crushing process and paving aggregate production

Aggregate production changes a natural or reclaimed feed into identified products. Each stage controls a different feature: crushing changes particle size and shape; screening separates sizes; washing removes specified unwanted fines; handling preserves the product. A plant diagram is useful because a defect observed at the stockpile may have begun several stages earlier.

The paving crew normally receives a finished product, but understanding its route helps them ask precise questions about segregation, contamination and supply. The material record should preserve source, product, test identity and destination.

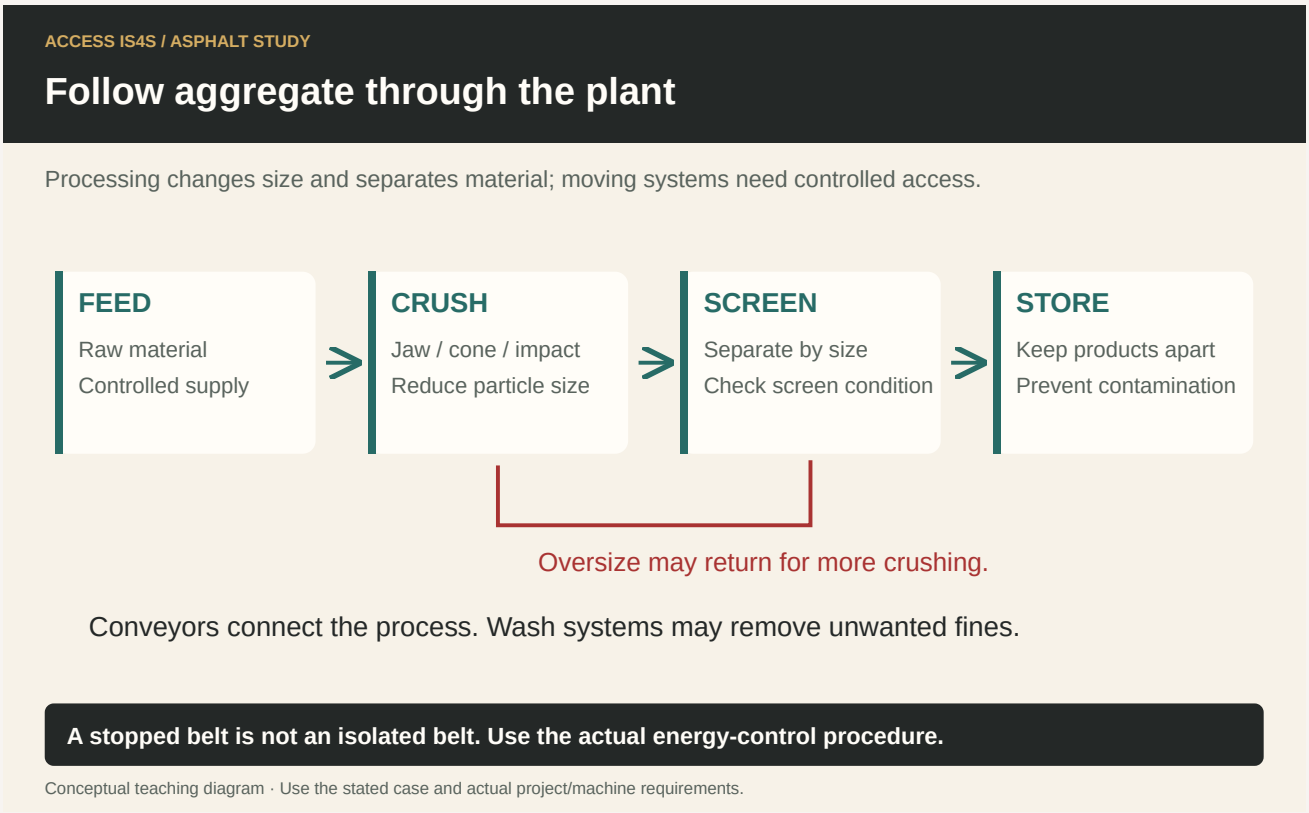
- Gravel pits, quarries
- Crushing equipment
- The screen
- Conveyors
- Feeders
- Explain stockpiling and handling
- Washing procedures
- Types of aggregate transportation

LEARNING STEP 1

Gravel pits, quarries

A **gravel pit** extracts deposits that can contain varying proportions of gravel, sand and fines. A **quarry** extracts rock that commonly needs breaking and crushing. Deposits vary within a site, so one attractive sample does not establish an entire source's suitability. Source approval, extraction permission, environmental controls and safe access are separate requirements. Dredged or recovered material likewise needs its authorized source and product assessment.

Follow aggregate through the plant



Note. ACCESS IS4S. (2026). Follow aggregate through the plant [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Crushing equipment

A **jaw crusher** reduces feed between crushing surfaces using compression. A **cone crusher** reduces material between a moving and a fixed crushing surface. An **impact crusher** uses impact to break material. Feed properties, desired product and the designed circuit influence selection and result. A learner identifies function and flow, not an operating gap or speed from a generic sketch. Wear and inappropriate feed can change shape, size distribution and equipment demand.

LEARNING STEP 3

The screen

A **screen** separates material through openings of stated sizes. Material that passes an opening follows one route; retained material follows another, which may return to crushing. A torn, blinded or incorrectly configured screen can change the product even when the crusher is operating normally. Read the screen/product relationship and the inspection/test evidence together. A screen opening description alone is not the full aggregate specification.

LEARNING STEP 4

Conveyors

A **conveyor** carries material between stages or to storage. Fixed, transfer and stockpiling arrangements have different purposes. Belt, idlers, pulleys, drive, guards and discharge point are parts to identify. Transfer height, direction and handling can affect segregation and dust. The carrying route must be distinguished from a safe pedestrian route; a stopped belt is not an authorized crossing or maintenance condition.

LEARNING STEP 5

Feeders

A **feeder** controls how material enters a process. Its job is more than moving the largest possible amount: a consistent, suitable feed helps the downstream circuit operate as intended. Irregular feed, bridging or incompatible material can disturb processing. Changes are handled through the plant's procedure; workers do not enter a hopper or reach into a feeder to correct the flow.

LEARNING STEP 6

Explain stockpiling and handling

Stockpiles keep identified products separate before use. Place and recover material using the approved handling arrangement to limit intermingling, segregation and contamination from the ground or adjacent products. Large particles can roll farther during pile formation, creating a different gradation at different locations. Maintain signs and records so a loader does not supply the wrong pile. Moisture and recovered-material variability also matter to later plant control.

LEARNING STEP 7

Washing procedures

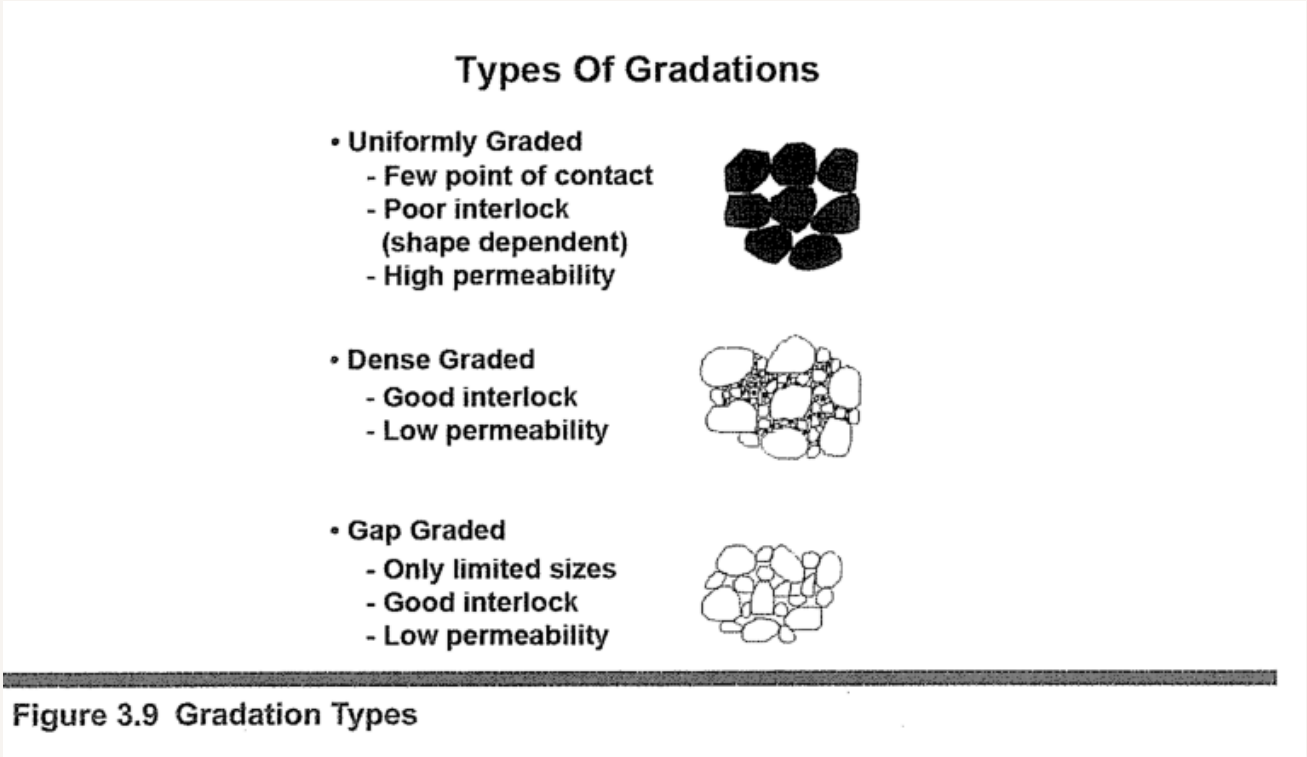
Washing separates unwanted fine material through a designed water-and-material process. It can improve cleanliness for the intended product, but also creates wash water and collected fines that need permitted management. Washing does not automatically make a product suitable for every use. Check the specified test results and the site's water/settling or recycling arrangement; uncontrolled discharge is not part of a legitimate material-improvement process.

LEARNING STEP 8

Types of aggregate transportation

Trucks, rail and barges can move aggregate between source, storage and plant. The selected transport method affects logistics and handling points, not the required product identity. Loads must remain traceable and protected against unacceptable contamination or segregation. Receiving checks compare the supplier, product, quantity and relevant test record with the order. A change of transport route or transloading point may introduce a new handling risk that belongs in the supply plan.

Gradation types



Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 3.9, p. 134; PDF p. 168). Faithful scan crop; original figure caption retained. Full source context.

Worked example · Find the process change

Fictional circuit: a screen sends its oversize back to a cone crusher. A damaged screen allows larger particles into the finished-product pile. The delivery ticket still carries the correct product name.

Think it through.

The name on the ticket has not verified the physical product. Identify affected production and stockpile records, isolate the questioned product through the quality process, inspect the screen safely and obtain the specified testing. Do not blend the questioned pile into accepted material to hide the difference. The process diagram points to where the change occurred and which records may be affected.

Try it and explain your thinking

Trace a product through feeder, jaw/cone/impact stage, screen, conveyor, stockpile and transport

Trace a product through feeder, jaw/cone/impact stage, screen, conveyor, stockpile and transport. Explain which stages change material and which preserve or separate it.

Skills you use: Reading, Communication.

A fictional delivery uses 12 truckloads of 18 t

A fictional delivery uses 12 truckloads of 18 t. Two loads are held after the screen fault is discovered. Calculate the recorded total and the quantity held, without calling the remainder accepted.

Skills you use: Numeracy, Problem solving, Writing.

Create a digital material-route record for the worked case

Create a digital material-route record for the worked case. Ask a partner to trace affected loads back to the pile and production interval; improve any missing identifiers.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Observe the actual process from authorized locations under site induction.	0 / 1 / 2	Evidence/help: _____
Assess identification of crushers, screens, feeders, conveyors, stockpiles and transport interfaces, plus the learner's explanation of product identity.	0 / 1 / 2	Evidence/help: _____
Sample handling and machine service require their own safe procedures and authorized personnel.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, C1, printed p. 37; source PDF pp. 39–39.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 3: Mineral Aggregates and Mixture Additives; source PDF pp. 143–192. Open publication.
- BC Ministry of Mining and Critical Minerals. (June 2026). Health, Safety and Reclamation Code for Mines in British Columbia. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 12: Tools, Machinery and Equipment. Open publication.

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Study chapter · Outline C2

Types and specifications of aggregates

An aggregate name is the beginning of a specification, not its complete meaning. The useful questions are: What particles are present? How do they interact? How do water and loading affect them? Which measured properties must the product meet? **Gradation** is the distribution of particle sizes; the properties of the blend are different from the properties of a single stone.

Aggregate processing

Characteristics

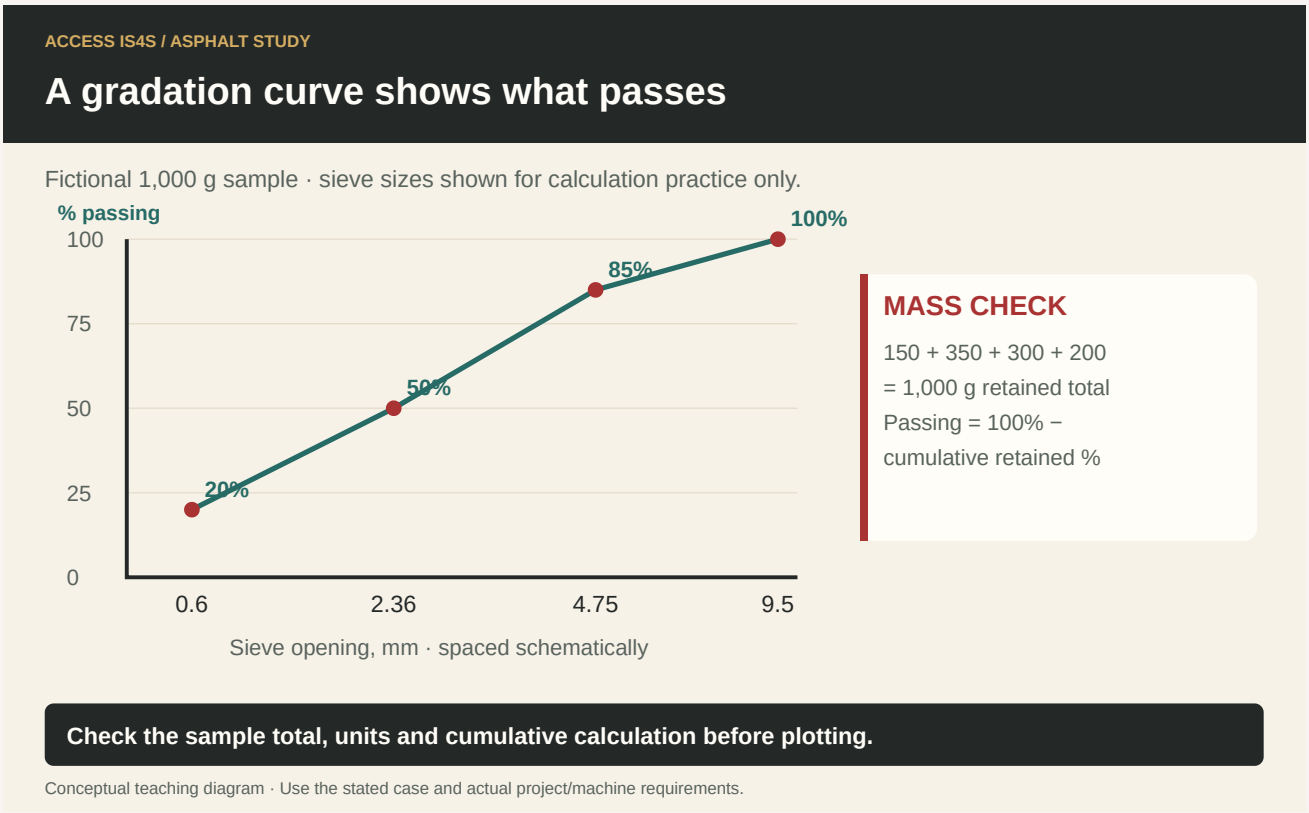
Common products

LEARNING STEP 1

Aggregate processing

Crushing, screening and washing perform different jobs. Crushing can alter size and particle shape; screening separates sizes; washing removes specified fine material. A product may pass through more than one stage. The process name does not prove the final result: test the identified product using the required method and compare it with the applicable specification. A change in source, wear or handling can affect results even when the nominal process remains the same.

A gradation curve shows what passes



Note. ACCESS IS4S. (2026). A gradation curve shows what passes [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Characteristics

Permeability describes the ability of a material arrangement to transmit water. **Load-bearing response** concerns how the structure supports loading. **Resistance to shearing** depends on particle interaction and other conditions. Particle shape, texture, grading, fines and moisture can influence these behaviours. An open-looking product is not automatically a designed drainage layer; the complete gradation, contamination and drainage arrangement matter.

Liquid limit and **plastic limit** characterize consistency of the applicable fine-grained material under specified tests. The **plasticity index** is liquid limit minus plastic limit. These are not moisture limits for every whole aggregate or a visual test of a pile. Gradation is normally recorded as cumulative percent passing identified sieves. Keep sieve size, units, test method and product/sample identity with the result.

LEARNING STEP 3

Common products

Pit run describes material taken from a deposit with limited processing; its properties can vary. **Screened road base**, including a nominal '3-inch minus' description, needs the specified grading and quality limits. **Bedding sand** is selected for the relevant pipe or bedding design. **Crushed road mulch** is a

regional/product term that must be resolved through its actual specification. **Asphalt aggregates** are selected and blended for the approved mixture.

Drain rock serves an intended drainage use with specified grading/cleanliness. **Chips** and **driveway chips** describe particular aggregate products whose intended application and grading must be established. **Reclaimed asphalt pavement (RAP)** contains recovered aggregate and aged binder; it is not interchangeable with clean virgin stone. **Concrete sand**, including material specified under ASTM C33, has its own requirements. **Stucco sand** is selected for a different mortar/render application. A shared word 'sand' does not authorize substitution among them.

Gradation types

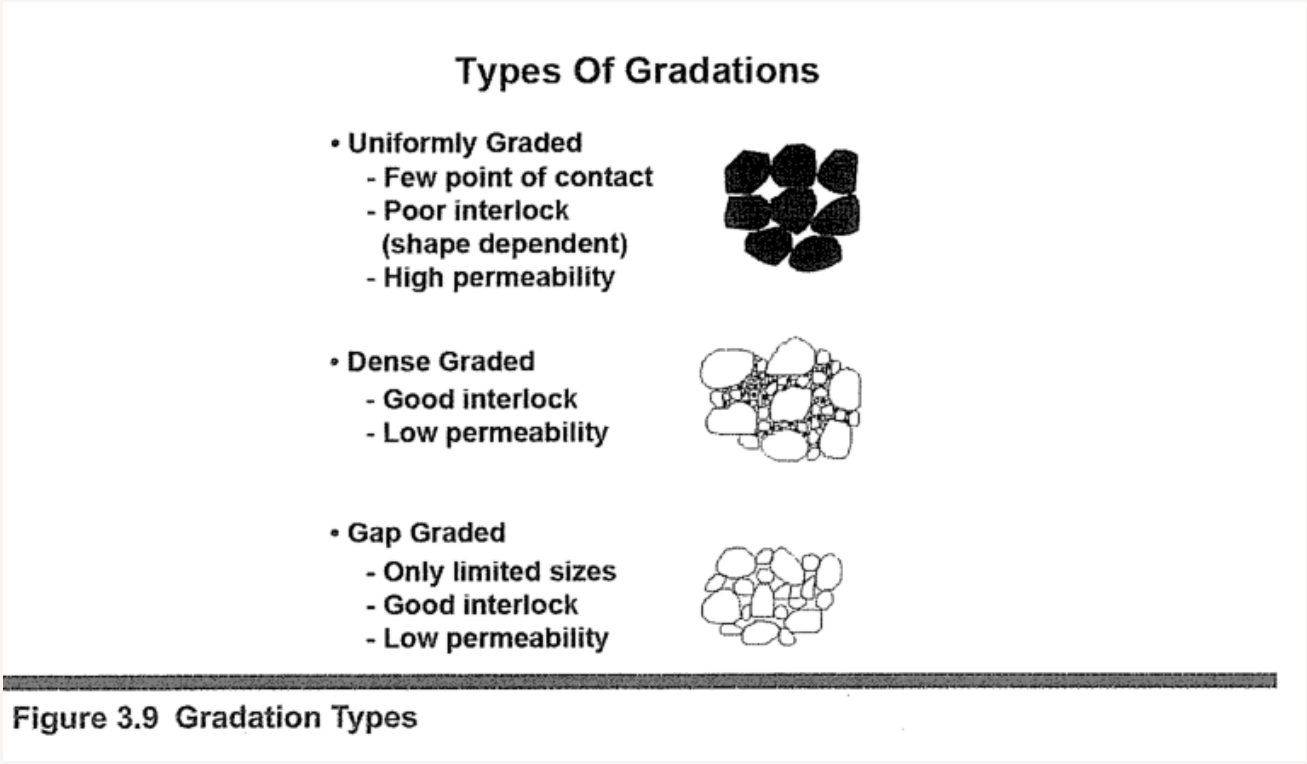


Figure 3.9 Gradation Types

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 3.9, p. 134; PDF p. 168). Faithful scan crop; original figure caption retained. Full source context.

Worked example · Read a sieve result

Fictional dry sample mass is 1,000 g. The measured retained masses on three successive sieves are 150 g, 350 g and 300 g; the pan holds 200 g. Use these values only to practise the arithmetic.

Think it through.

The masses total 1,000 g. Cumulative retained percentages are 15%, 50% and 80%; cumulative passing values are 85%, 50% and 20%. The pan fraction is 20%. Whether the product conforms cannot be decided without sieve sizes, specified method, product identity and acceptance limits.

Try it and explain your thinking

Make a product selection table for bedding, drainage, asphalt mixture and concrete sand

Make a product selection table for bedding, drainage, asphalt mixture and concrete sand. Name the information needed before accepting a substitution.

Skills you use: Reading, Writing, Problem solving.

Recalculate the example if 20 g is missing from the total

Recalculate the example if 20 g is missing from the total. Then calculate PI for fictional LL = 32 and PL = 21. Explain the limits of both results.

Skills you use: Numeracy, Communication.

Plot the supplied cumulative passing values with clearly labelled sieve positions

Plot the supplied cumulative passing values with clearly labelled sieve positions. Save the chart, let a partner find the missing real sieve sizes, and revise its title to avoid implying acceptance.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use labelled samples and the approved test method under supervision; do not identify a product by colour alone.	0 / 1 / 2	Evidence/help: _____
Observe correct terminology, sample/product traceability and interpretation of a sieve table.	0 / 1 / 2	Evidence/help: _____
Laboratory competence and approval of a source/product require the applicable qualified testing and decision process.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, C2, printed p. 38; source PDF pp. 40–40.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 3: Mineral Aggregates and Mixture Additives; source PDF pp. 143–192. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.

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Study chapter · Outline C3

Asphalt plant operation

Both drum and batch plants turn specified inputs into a controlled mixture, but they organize the flow differently. Learn the path of aggregate, binder, recovered material where permitted, collected fines and finished mixture. A material-flow drawing is not a pipe-and-instrumentation diagram: it explains relationships without supplying the plant's operating settings.

The operation of a drum plant

The operation of a batch plant

LEARNING STEP 1

The operation of a drum plant

In a continuous drum-mix arrangement, **cold-feed bins** supply identified aggregate fractions. Feeders and conveyors proportion and carry material through the designed drying/mixing system. Binder and other permitted inputs are introduced through controlled systems. Drum layouts differ, including where drying and mixing occur; do not infer one universal arrangement from the word 'drum'. Moisture in the aggregate affects drying demand and production control.

Finished mixture moves to the designed storage/surge system and **loadout**. A **storage silo** provides a controlled material-handling stage; it does not remove the need to preserve mix identity, temperature conditions and acceptable storage duration. The load record connects the product and time to the truck. Dust collection, burner performance and recovered-fines control interact with production and environmental requirements. A change in one feed must be managed through the approved mixture/plant process.

Batch or continuous?

ACCESS IS4S · ASPHALT FOUNDATIONS

Batch or continuous?

Classify the mixing process, not the presence of one piece of equipment.

BATCH Separate weighed batches

Cold-feed bins → Dry and heat → Screen + hot bins → Weigh a batch → Mix with binder → Store / load out

A batch plant can have a drying drum. The batch mixer defines the process.

DRUM / CONTINUOUS

Proportion at cold-feed bins → Measure the material flow → Dry + mix with metered binder → Continuous mix to storage

Some facilities use a separate dryer and continuous mixer.

Simplified material routes · additives, recycling and detailed controls require their own records.

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Original instructional adaptation by ACCESS IS4S (2026). Converted equipment drawings into two simplified process lanes; omitted settings and detailed control systems. Source locations: [{"figure": "7.2", "printed_page": 310, "pdf_page": 344}, {"figure": "7.4", "printed_page": 312, "pdf_page": 346}, {"section": "7.1.1–7.1.2", "printed_pages": [309, 311, 312], "pdf_pages": [343, 345, 346]}]. Original contributors: Terex Roadbuilding; National Highway Institute (NHI) Full source context.

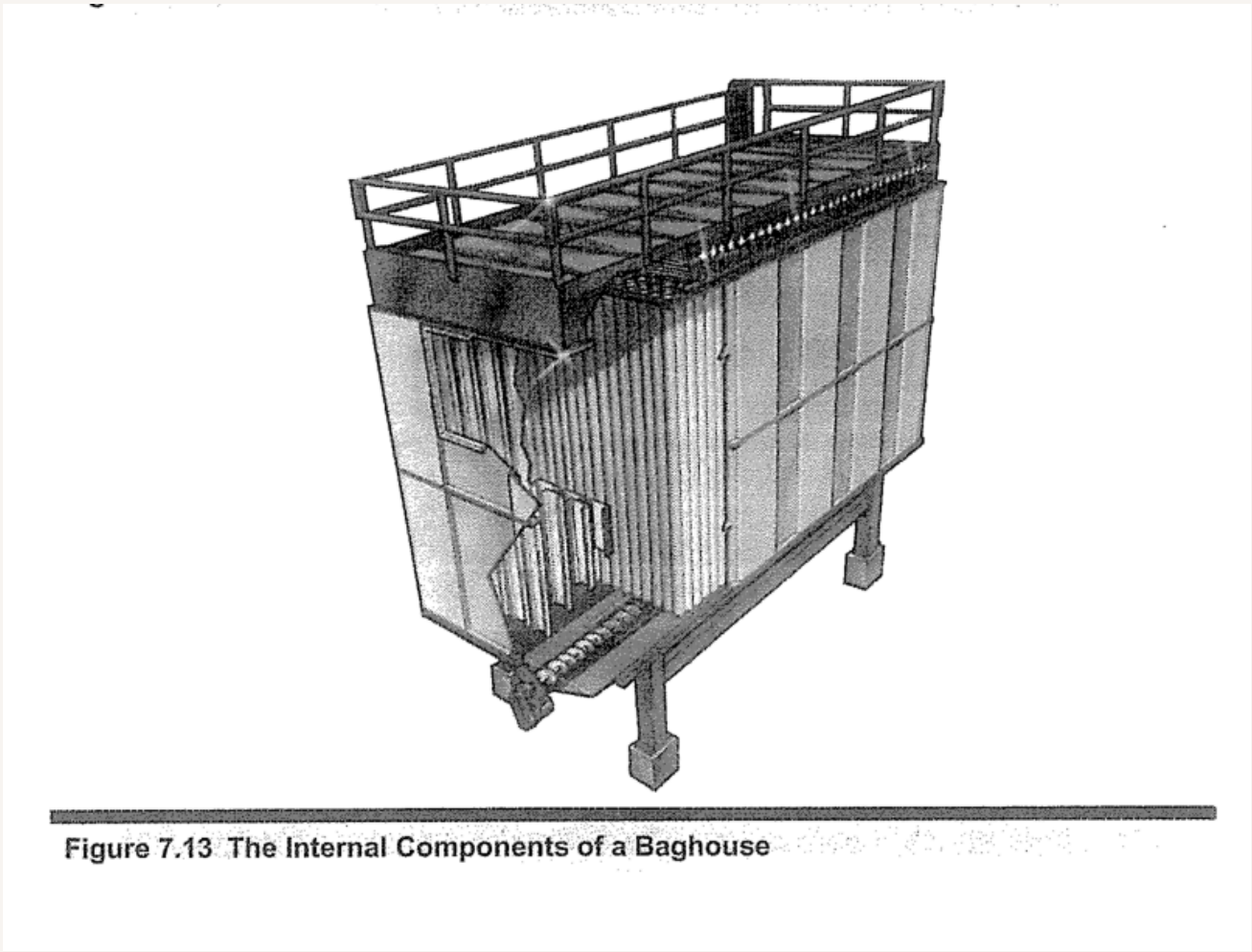
LEARNING STEP 2

The operation of a batch plant

In a conventional batch arrangement, cold feeds send aggregate to drying/heating. A **hot stone elevator** lifts heated aggregate to the **screen deck**, where fractions are separated into **hot bins**. **Weigh bins** measure the selected fractions for a batch. The **pug mill** mixes the measured materials with the specified binder and other permitted inputs. A **slat conveyor** or other designed transfer moves the product toward storage/loadout.

The batch identity helps relate proportioning to a production record. It does not guarantee that every batch meets the formula: incorrect bin contents, screen condition, scales, moisture assumptions or binder measurement can affect the result. Quality control checks the process and tested product. Operators follow the actual plant's safe controls, interlocks and procedures; learners should be able to trace an input and explain what evidence would reveal a change.

Internal components of a baghouse



Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 7.13, p. 324; PDF p. 358). Faithful scan crop; original figure caption retained. Original image contributor: Terex Roadbuilding; credit register, PDF pp. 803–805. Full source context.

Worked example · One product, two process records

Fictional batch card: 950 kg aggregate plus 50 kg binder, with no other mass included. Fictional continuous record: 19,000 kg aggregate plus 1,000 kg binder over an interval. Compare binder mass percentage on a total-mixture basis.

Think it through.

Batch total = 1,000 kg; binder = 5%. Continuous total = 20,000 kg; binder = 5%. The same arithmetic ratio does not establish matching gradation, moisture correction, binder grade, temperature or conformity. The records describe different production organizations; both require the approved formula and checks.

Try it and explain your thinking

Explain and apply the idea

Trace both plant diagrams and explain hot elevator, screen deck, hot bin, weigh bin, pug mill, slat conveyor, silo and loadout.

Skills you use: Reading, Communication.

Recalculate the batch if binder is 52 kg while aggregate remains 950 kg

Recalculate the batch if binder is 52 kg while aggregate remains 950 kg. State the denominator and why the result alone cannot approve the product.

Skills you use: Numeracy, Writing, Problem solving.

Create a saved side-by-side production record

Create a saved side-by-side production record. A partner introduces a bin-label mismatch; identify affected records and revise the release status.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use the actual plant process drawing and authorized observation area.	0 / 1 / 2	Evidence/help: _____
Assess component/function identification, input/output traceability and recognition of a changed production condition.	0 / 1 / 2	Evidence/help: _____
Plant operation, scale calibration and maintenance are separate authorized practical tasks.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, C3, printed p. 39; source PDF pp. 41–41.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 7: Manufacture of Hot Mix Asphalt; source PDF pp. 341–390. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Asphalt Plant Regulation, B.C. Reg. 217/97. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 12: Tools, Machinery and Equipment. Open publication.

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Study chapter · Outline C4

Asphalt products and uses

A road requirement has an issuer, edition, project adoption and decision authority. A contractor, designer, owner and regulator may each issue documents, but their documents answer different questions. Read the contract register before choosing a clause from a familiar book. The same surface material can appear on a highway, a municipal street and private property under different project requirements.

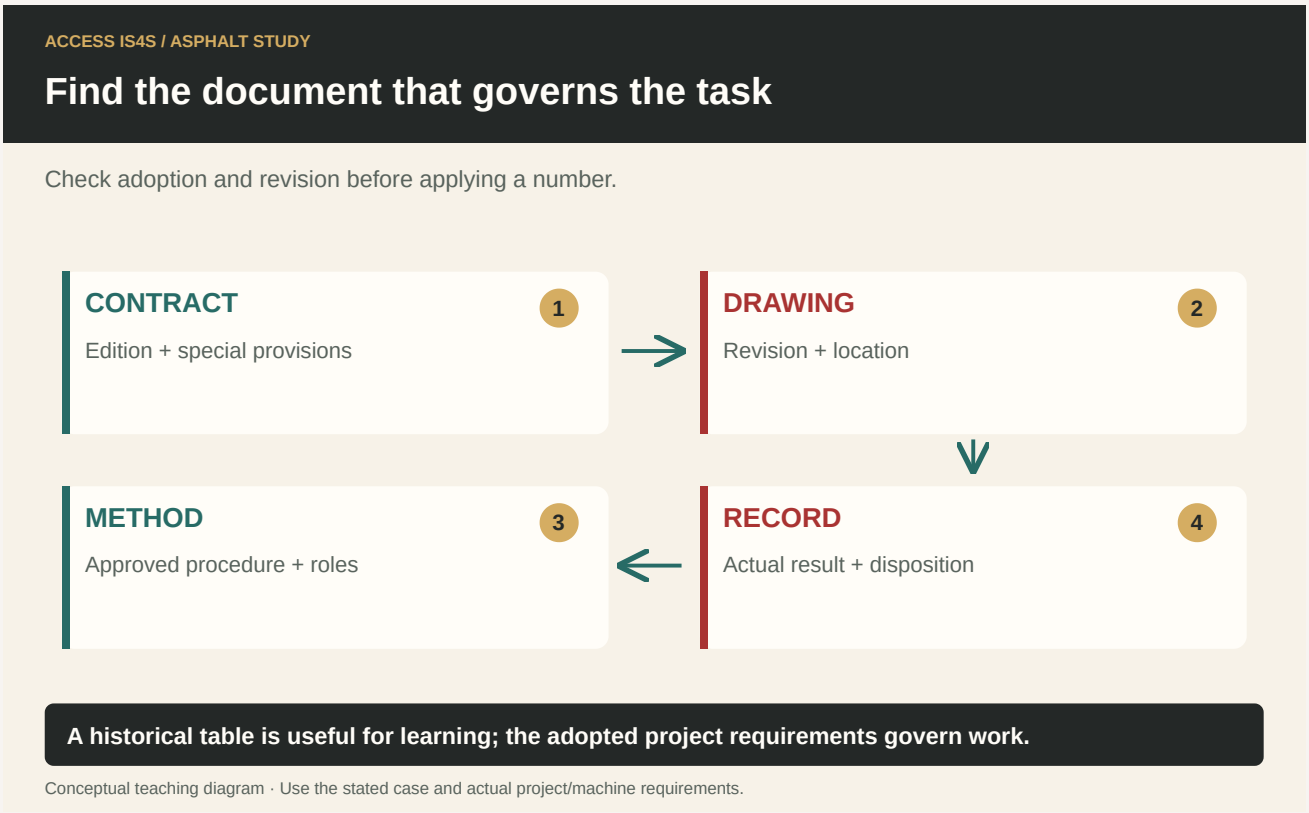
- The key elements of the MMCD
- The role of the MOT
- The role of the DOT
- Engineer Specs
- Role of Architect
- Other asphalt and paving materials

LEARNING STEP 1

The key elements of the MMCD

MMCD means Master Municipal Construction Documents. The published contract documents provide standardized municipal construction specifications, drawings and related contract material. A project may adopt an edition and add supplementary conditions or specifications. Obtain the actual adopted documents and changes; the title 'municipal work' does not establish every MMCD clause or edition. Record the governing project reference, not a guessed requirement from another job.

Find the document that governs the task



Note. ACCESS IS4S. (2026). Find the document that governs the task [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

The role of the MOT

The provincial transportation ministry's role is established by the project and applicable law/contract. Distinguish contractor **quality control** from owner-side **quality assurance**, audit and acceptance activities as defined in the contract. A contractor's passing internal check is not automatically the owner's acceptance. Conversely, an audit sample does not remove the contractor's duty to control its own process. Current ministry naming can change; identify the actual issuer and authorized representative.

LEARNING STEP 3

The role of the DOT

DOT is commonly used for a Department of Transportation in other jurisdictions. It does not identify one universal Canadian asphalt standard. The supplied 2006 B.C. Design Build Standard Specifications is a specific historical document, not a generic federal transport regulation. When an old note says 'DOT standard', identify its jurisdiction, full title, edition and contractual adoption before using it. This prevents a familiar acronym from becoming an unsupported requirement.

LEARNING STEP 4

Engineer Specs

An engineer’s project specifications translate the design into required materials, workmanship, inspection and acceptance criteria. Such specifications are used in public as well as private work. Drawings and specifications must be read together, using the project’s hierarchy and change process. A conflict is referred to the authorized design/contract role; the crew does not choose whichever version gives the easiest quantity.

LEARNING STEP 5

Role of Architect

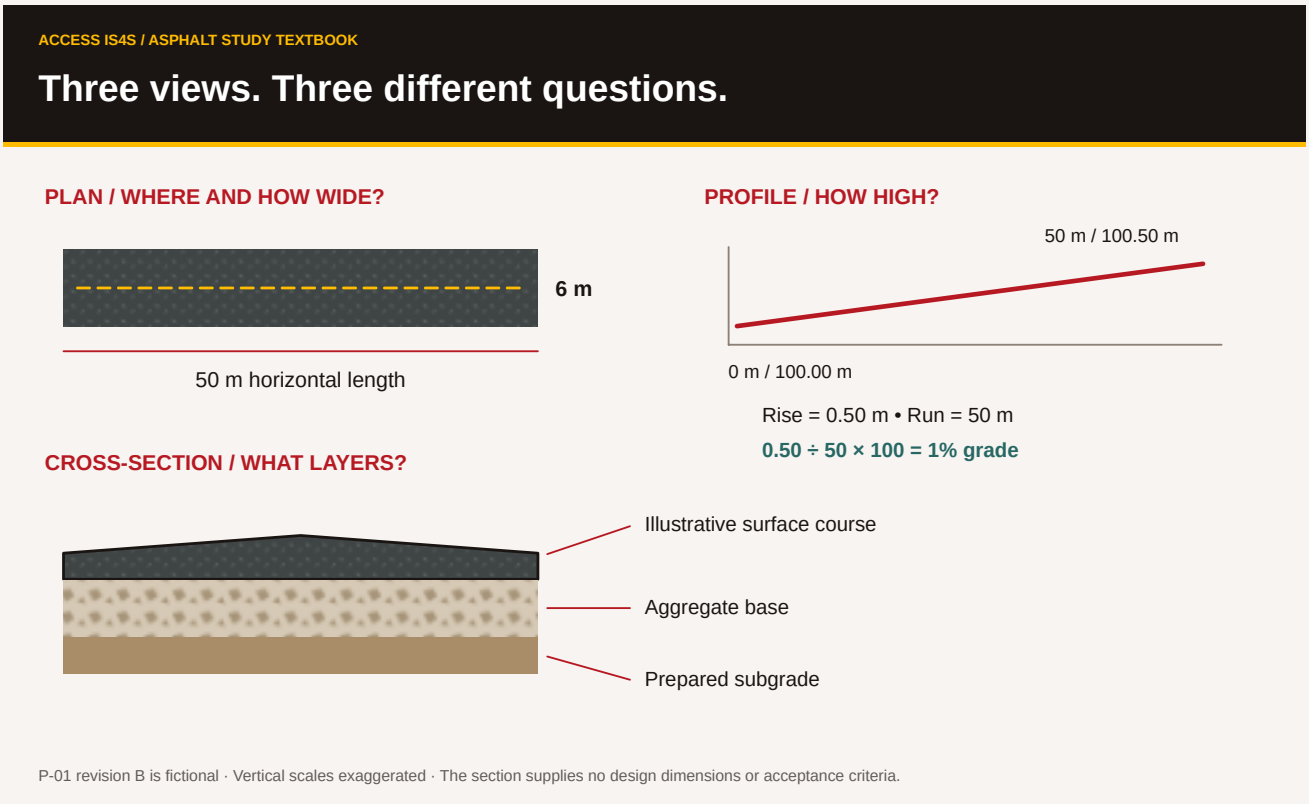
An architect may coordinate the site and building interface, accessibility, appearance and other project requirements through the design team. The actual division of responsibility is project-specific. Architectural drawings can identify finishes or interfaces while civil drawings establish drainage and pavement details. A worker must know which record and responsible designer address the question, and retain an authorized response when the work changes.

LEARNING STEP 6

Other asphalt and paving materials

Paving blocks/stones, asphalt, concrete and permeable pavement systems have different structures, construction methods and maintenance needs. 'Green road' is a claim or design objective that needs defined performance and evidence. A substitution affects more than appearance: loading, drainage, access, joints, durability and approved quantities may change. Compare the required service and obtain the authorized design decision before substituting a product.

Three views. Three different questions.



Note. ACCESS IS4S. (2026). Three views. Three different questions. [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · Resolve the document conflict

Fictional register lists drawing P-04 revision C and specification SP-07 revision 2. A delivery note refers to an older project's MMCD detail. The crew is asked to change the surface material at the entrance.

Think it through.

The delivery note does not establish the governing design. Identify the current project documents, describe the requested change and obtain the authorized design/contract response. Keep the reason and affected drawing/specification revisions together. Do not treat an acronym or a supplier recommendation as approval.

Try it and explain your thinking

Explain and apply the idea

Create a five-column authority map: owner, designer, contractor QC, owner QA/acceptance and regulator. Place MMCD/ministry/project documents in their actual roles.

Skills you use: Reading, Writing, Problem solving.

Calculate and check the result

Write a clarification request for the worked case that names the conflict, location, current versions and decision needed.

Skills you use: Writing, Communication.

Communicate and verify the work

Build a digital revision register and peer-test whether another learner can find the current entrance requirements. Add a new response and show the superseded record without deleting its history.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use a permitted project document set or the complete fictional case.	0 / 1 / 2	Evidence/help: _____
Assess correct document selection, interpretation and referral of an unresolved design conflict.	0 / 1 / 2	Evidence/help: _____
No classroom interpretation changes a real contract or design.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, C4, printed p. 40; source PDF pp. 42–42.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 2: Petroleum Asphalt; source PDF pp. 55–142. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 15: Special Uses of Asphalt and HMA; source PDF pp. 737–802. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.
- Master Municipal Construction Documents Association. (Web snapshot: October 30, 2026). Supplementary Documents. Open publication.

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Study chapter · Outline D1

Environmentally friendly properties

An environmental claim is useful when it names the material, exposure conditions, service provided and evidence. Asphalt can be recovered and used again, but 'recyclable' is not the same as 'recycled here', and neither word proves the lowest overall impact. The learning goal is a defensible comparison that respects the people, land and water affected by a project.

Define Green properties of asphalt

LEARNING STEP 1

Define Green properties of asphalt

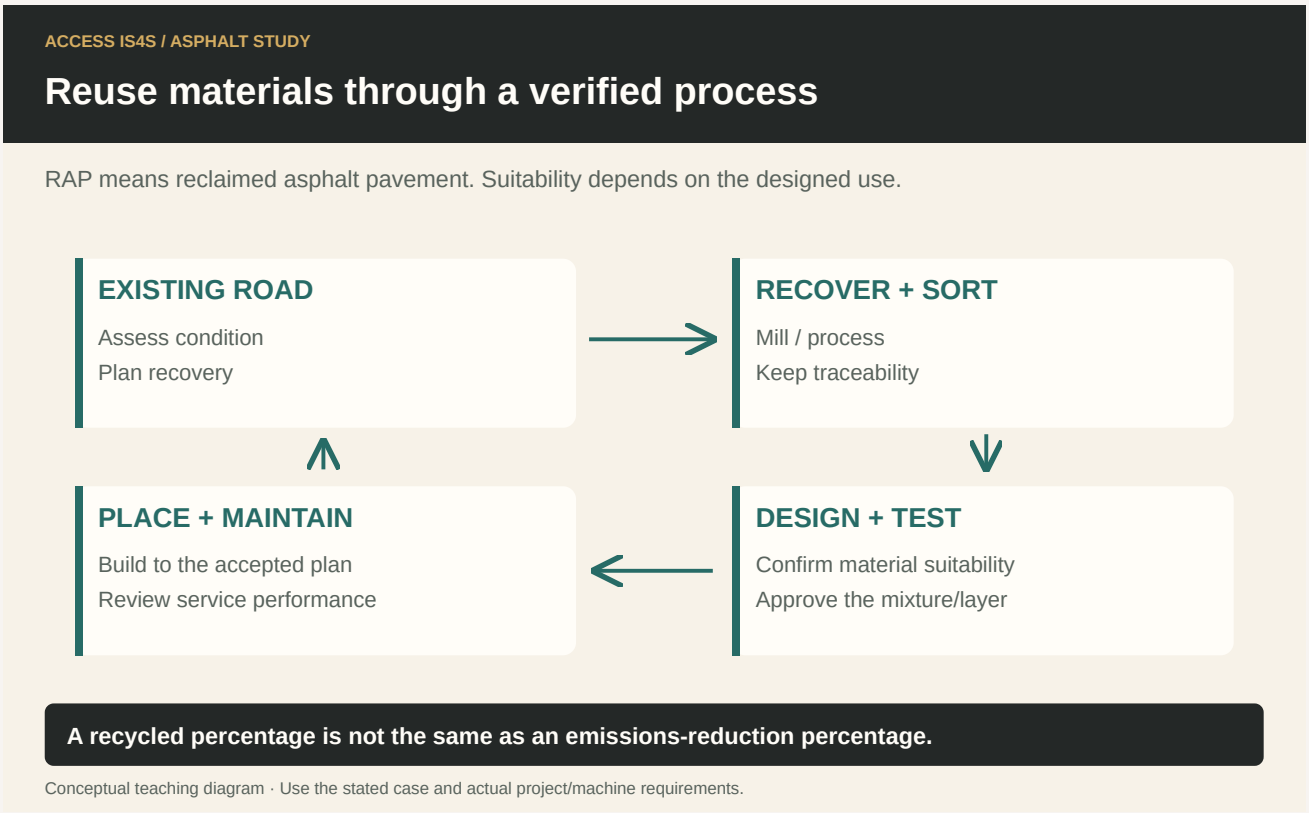
Reclaimed asphalt pavement (RAP) contains recovered aggregate and aged binder. Processing and an approved mix or other permitted use can recover value from those materials. The receiving process must account for variability, contamination, binder properties and the required final performance. A recycling pathway has collection, processing, transport and production effects of its own.

Avoid blanket claims that all installed asphalt is 'non-toxic' or 'does not leach' under every condition. Product composition, contamination, exposure, receiving environment and applicable criteria determine what evidence is needed. An intact pavement, hot production process, milling dust and a stored pile are different exposure situations. A claim about one does not automatically apply to the others.

A fair comparison defines the same required service, area/loading, service period and maintenance assumptions. Include production, transport, construction, use/maintenance and end-of-life within the stated boundary. A lower production impact per tonne can be outweighed by different quantities, travel or service life. Environmental decisions can also involve water, noise, land, access and local priorities that a single emissions number does not capture.

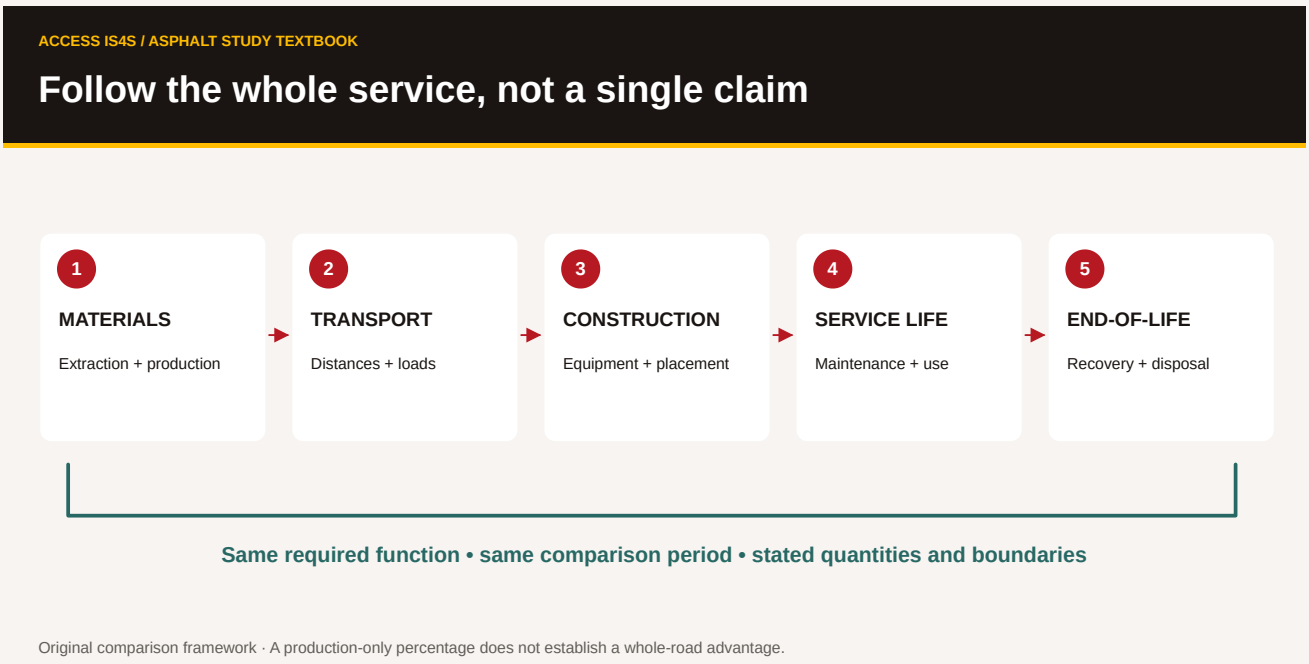
In B.C., meaningful local engagement identifies the appropriate people and decision processes; a class cannot supply a Nation’s permission or knowledge. Use public/fictional evidence for the exercise and invite contributions only by choice. A careful conclusion can acknowledge a documented reuse pathway while asking for the data needed to compare actual options.

Reuse materials through a verified process



Note. ACCESS IS4S. (2026). Reuse materials through a verified process [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Follow the whole service, not a single claim



Note. ACCESS IS4S. (2026). Follow the whole service, not a single claim [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · Recycled proportion is not a full impact score

Fictional material record: a 240 t proposed mix contains 48 t of approved RAP. A sales note calls it “20% recycled, therefore 20% lower whole-life emissions.” No emissions or service-life data is supplied.

Think it through.

48/240 × 100 = 20% by mass in this record. That establishes the stated proportion only. Whole-life emissions need comparable quantities, processing/production, transport, construction, maintenance and end-of-life evidence. The proposed reduction is not demonstrated by the recycled percentage.

Try it and explain your thinking

Explain and apply the idea

Rewrite the sales note into one supported statement and three questions needed for a whole-life comparison.

Skills you use: Reading, Writing, Problem solving.

Calculate the RAP mass for a second fictional 300 t mix at 15%

Calculate the RAP mass for a second fictional 300 t mix at 15%. Explain why comparing 48 t with the second mass does not rank environmental performance.

Skills you use: Numeracy, Communication.

Create a visual comparison brief with explicit boundaries

Create a visual comparison brief with explicit boundaries. Test it with a partner who asks about runoff or community access, then revise and save it.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Assess evidence interpretation using documented or fictional product records.	0 / 1 / 2	Evidence/help: _____
Check that the learner distinguishes recovery potential, actual reuse and verified environmental outcomes.	0 / 1 / 2	Evidence/help: _____
Environmental approvals and material-use decisions remain with the appropriate qualified/authorized processes.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, D1, printed p. 41; source PDF pp. 43–43.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 13: Rehabilitation Methods for Asphalt Pavements; source PDF pp. 653–710. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Asphalt Plant Regulation, B.C. Reg. 217/97. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Report a Spill. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline D2

Use of storage areas

Storage keeps the right material identifiable and available while controlling release, incompatibility and exposure. A storage plan includes the container/tank, surrounding area, access, inspection, emergency response and destination for waste. A tidy-looking tank is not proof that its containment or product identity is adequate.

- Type 1 storage
- Type 2 storage
- Use of recycled materials in hot mix asphalt
- Use of recycled materials in asphalt plant burners

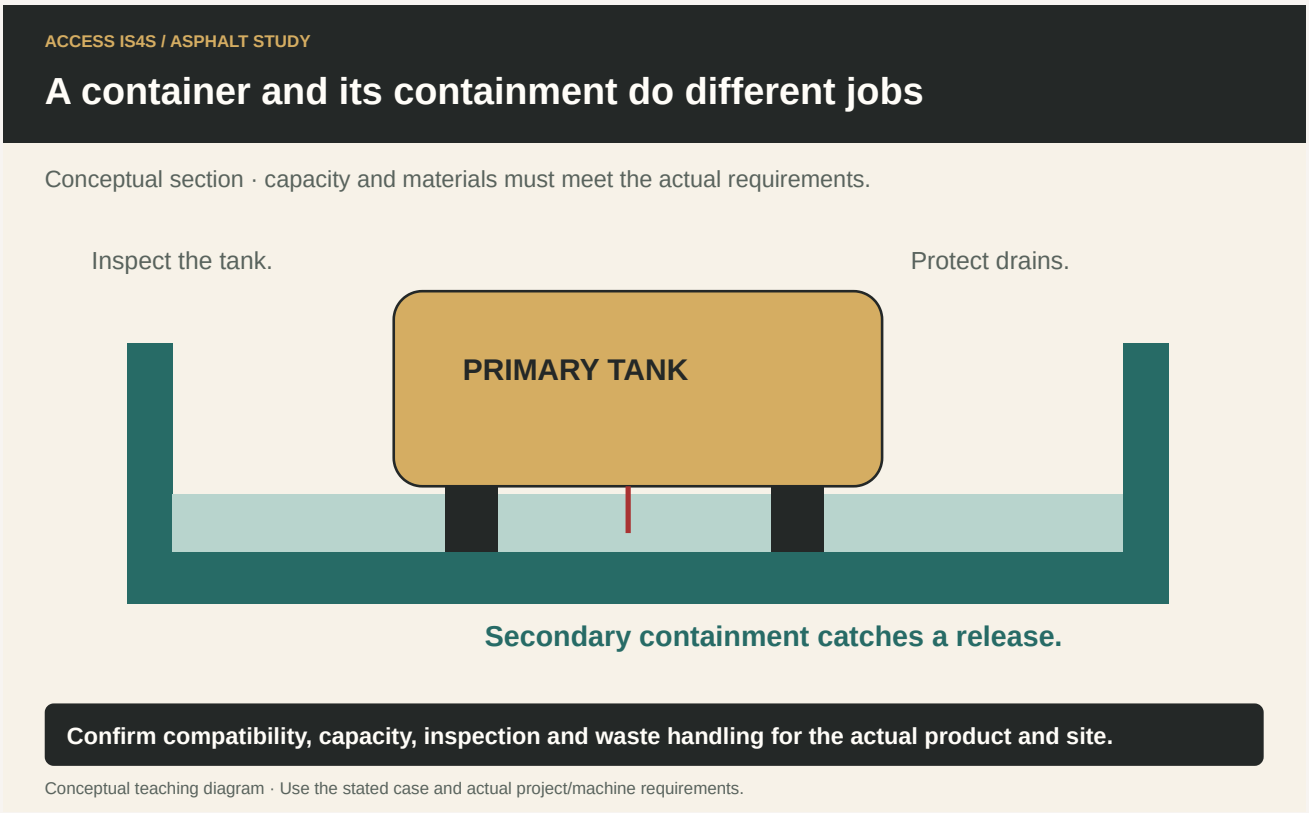
LEARNING STEP 1

Type 1 storage

For this lesson, **Type 1** describes a primary storage container with a separate spill-containment arrangement, such as a tank within a designed containment area. It is a course grouping, not a universal legal classification. Fuels, primers, emulsions such as an identified SS-1 product, asphalt binder and waste streams require their actual SDS, compatibility, heating/handling and regulatory arrangements.

Identify the product and hazards before deciding which area is suitable. Natural gas service has its own engineered supply and safety requirements; it is not handled as an ordinary open liquid tank. Containment must remain effective, including when rainwater or other material occupies space. Drainage and disposal of collected liquid follow the approved procedure; opening a drain to clear capacity can release contamination.

A container and its containment do different jobs



Note. ACCESS IS4S. (2026). A container and its containment do different jobs [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Type 2 storage

Type 2 here describes a self-contained storage arrangement, such as equipment with integral secondary containment, where its design provides that function. The word self-contained does not mean leak-proof, maintenance-free or automatically suitable for every fuel or liquid asphalt product. Verify the actual design, product compatibility, capacity, inspection and permitted location.

A biodiesel blend can have different handling/storage considerations from another fuel. A heated binder storage system introduces thermal and other hazards. Labels, isolation, overfill prevention, transfer connections and emergency access must match the actual installation. Read the tank/assembly instructions and product records together.

LEARNING STEP 3

Use of recycled materials in hot mix asphalt

Using RAP in hot mix can reduce demand for some new material and may offer cost benefits, but the outcome depends on processing, transport, consistency, binder contribution and required performance. Mobile recycling changes where processing occurs and may change traffic, emissions and material-control

arrangements. Identify what is being recycled and into which approved product; do not treat every recovered pile as acceptable feed.

Keep recovered material segregated by the required source/type and protect it from unacceptable contamination. The quality record follows it through processing and blending. A low purchase price alone is not the cost of a compliant finished mixture.

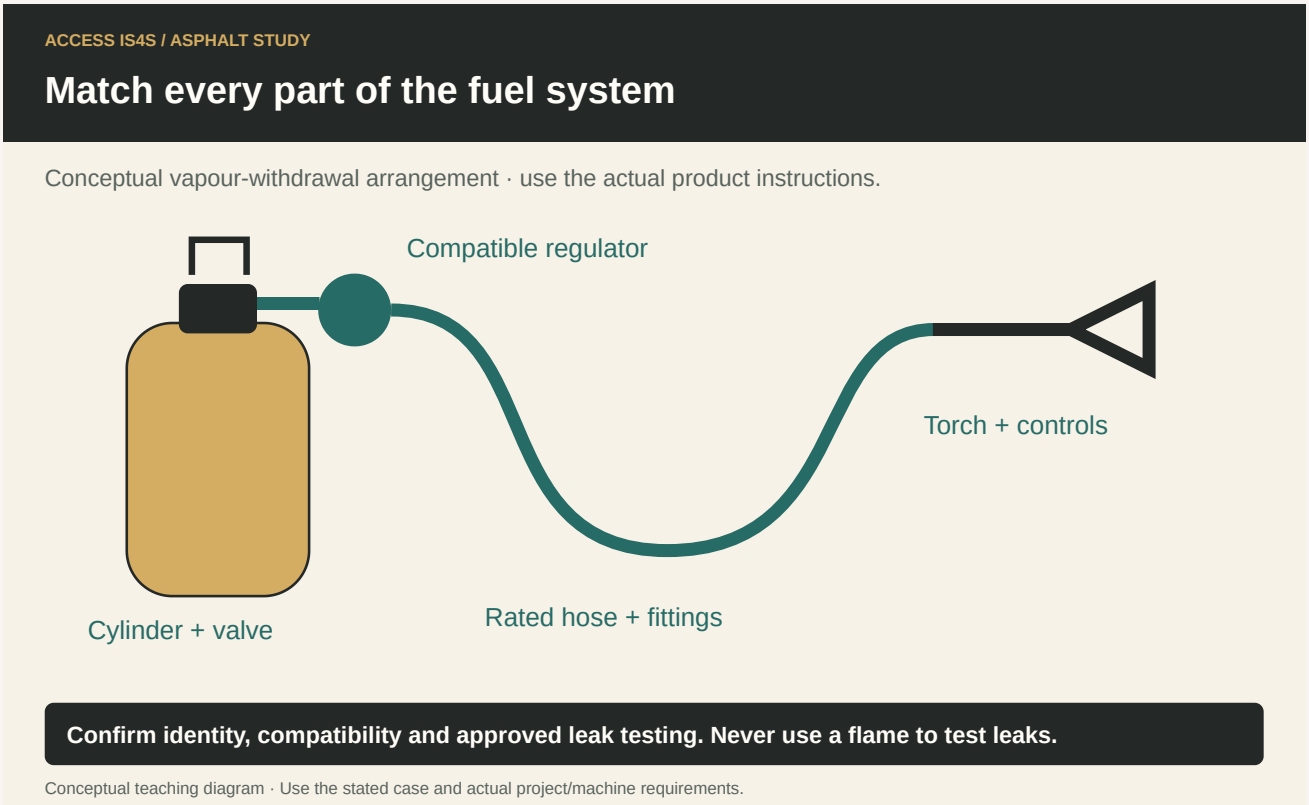
LEARNING STEP 4

Use of recycled materials in asphalt plant burners

A plant burner must use only fuel permitted by its design and applicable environmental authorization. Recovered fuel or other recycled material is not automatically acceptable because it burns. Composition, contamination, handling, burner suitability, combustion control and emissions requirements matter. Heating temperature is established by the approved process and equipment, not by a generic classroom number.

Storage/transfer for a permitted recovered fuel needs the actual SDS and emergency arrangements. A change in fuel can affect burner performance and emissions and must be controlled through the authorized process. Do not conduct a classroom trial with an unidentified waste fuel.

Match every part of the fuel system



Note. ACCESS IS4S. (2026). Match every part of the fuel system [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Containment is a working system

Fictional inspection record: the containment area is intact, but its drain is open and collected rainwater has not been assessed for contamination. A delivery is due in 30 minutes.

Think it through.

The record does not establish effective containment for the transfer. Have the responsible person implement the storage/transfer procedure, address the open drain and assess the collected liquid before release or transfer. Delivery timing does not change the product or environmental requirements. Record the condition and resolution rather than ticking “tank intact” as the whole inspection.

Try it and explain your thinking

Draw the difference between separate containment and an integral self-contained arrangement

Draw the difference between separate containment and an integral self-contained arrangement. Identify five checks both still require.

Skills you use: Reading, Communication.

Calculate and check the result

Fictional cost exercise: virgin material supply costs $8,000; recovered feed costs $5,000 plus $2,400 processing and $900 extra transport. Compare only these stated costs.

Skills you use: Numeracy, Writing, Problem solving.

Create a digital storage inspection record and peer-test it with the open-drain case

Create a digital storage inspection record and peer-test it with the open-drain case. Revise it to distinguish observation, action, responsible person and verified closure.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use the actual installation, SDS, transfer plan and permitted observation area.	0 / 1 / 2	Evidence/help: _____
Assess product identification, containment/compatibility questions and reporting of a defect.	0 / 1 / 2	Evidence/help: _____
Live fuel/binder transfer and burner changes require separate authorized instruction and controls.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, D2, printed p. 42; source PDF pp. 44–44.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 7: Manufacture of Hot Mix Asphalt; source PDF pp. 341–390. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 5: Chemical and Biological Substances. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Report a Spill. Open publication.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Chemical Profiles: Propane. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline D3

Asphalt plant and placement emissions

Emissions are releases from a process into the environment; worker exposure is contact with a hazard at the workplace. They are related but are not measured or regulated in exactly the same way. A stack result does not establish a worker's breathing-zone exposure, and a workplace mask does not establish compliance with an environmental permit.

- Discuss role of Metro Vancouver (formerly GVRD) in enforcement and regulation of emissions
- Role of MOE in establishing regulations
- Use of Bag house
- Emission controls

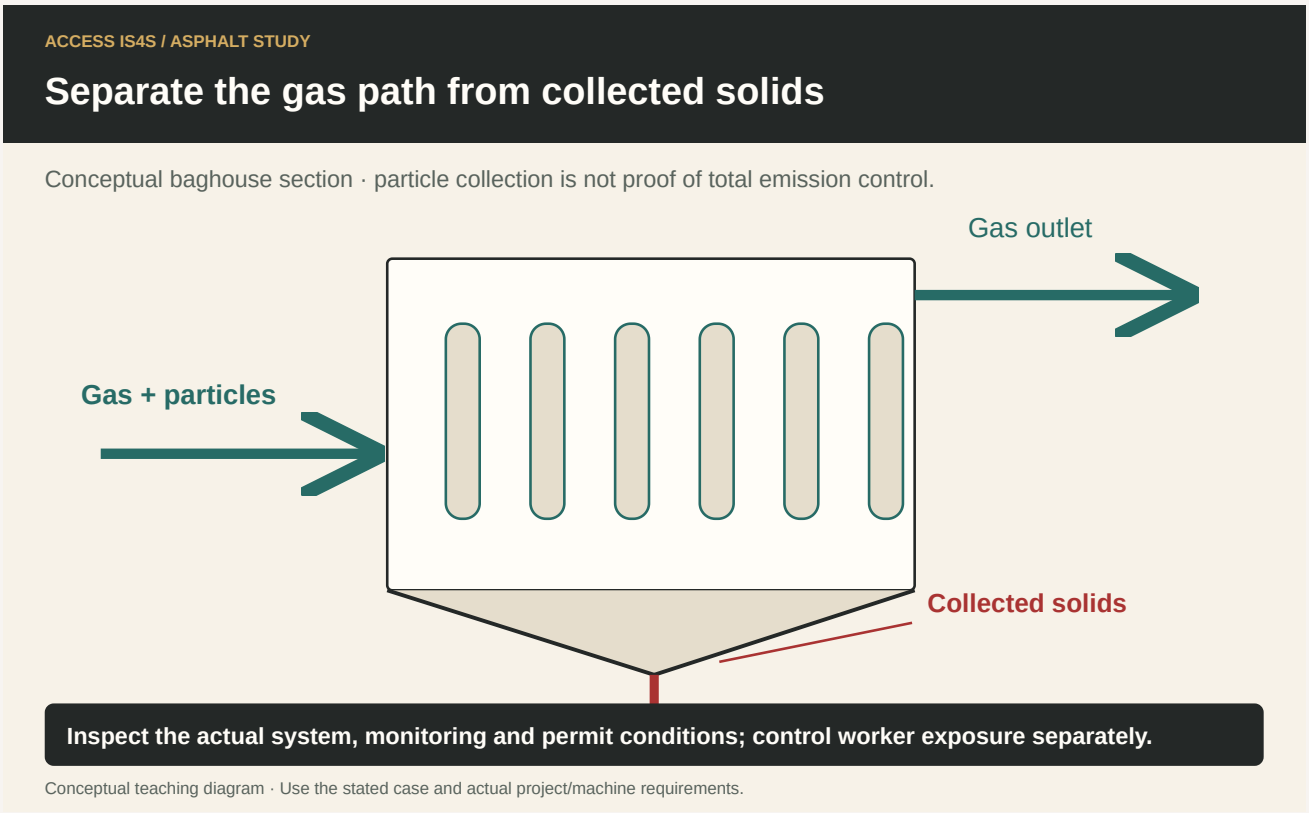
LEARNING STEP 1

Discuss role of Metro Vancouver (formerly GVRD) in enforcement and regulation of emissions

Metro Vancouver has a regional air-quality regulatory program. Determine whether the plant or activity is within that jurisdiction and which permit, bylaw or authorization applies. The retained B.C. Asphalt Plant Regulation explicitly states that it does not apply in the Metro Vancouver Regional District. Do not carry an outside-region authorization into the region by assumption.

The practical record identifies the actual site, operator, authorization, operating conditions, monitoring/reporting requirements and responsible contacts. A historical count of member communities is not a compliance test. Consult the current regional program and actual permit when preparing the work.

Separate the gas path from collected solids



Note. ACCESS IS4S. (2026). Separate the gas path from collected solids [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Role of MOE in establishing regulations

Outside Metro Vancouver, the B.C. Asphalt Plant Regulation establishes requirements for the covered hot-mix and hot-in-place recycling plants, with applicable registration, emission, operating and reporting provisions. Its text also addresses additional municipal/regional requirements. Identify the actual activity and location before selecting the rule. Provincial ministry names and responsibilities can change; the legal instrument and current authorized contact must be recorded.

An environmental authorization is different from the workplace exposure-control program. Both can be relevant to the same plant. Keep monitoring data and reports associated with their method, location, interval and required interpretation.

LEARNING STEP 3

Use of Bag house

A **baghouse** is a fabric-filter dust collection system. In the MS-4 concept, dust-laden gas passes through the filtering arrangement, particles collect and cleaning releases collected material to a hopper. The gas and collected-solids paths are different. Primary collection may remove larger particles before secondary collection. The actual plant arrangement determines how fines are returned or removed.

Filter condition, seals, airflow and the designed cleaning/collection system affect performance. A visible change, alarm or damaged component needs the operator's prescribed response. Do not assume a baghouse removes every gaseous pollutant: its principal dust-control function does not establish control of all emissions.

LEARNING STEP 4

Emission controls

Burner operation, aggregate moisture, production rate, material temperature, handling and dust-control performance can influence emissions. Heating asphalt excessively or using an unsuitable fuel can create problems; actual limits and operating settings come from the approved plant process and authorization. Placement work also needs exposure controls suited to the product and conditions.

Record observations without diagnosing the entire process from colour or odour alone. An unusual visible plume can prompt the required response and investigation, but it is not a measured concentration. Keep the operating interval and material/fuel identity so the responsible people can relate the observation to production and monitoring records.

Internal components of a baghouse



Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 7.13, p. 324; PDF p. 358). Faithful scan crop; original figure caption retained. Original image contributor: Terex Roadbuilding; credit register, PDF pp. 803–805. Full source context.

Worked example · Two measurements, different questions

Fictional records show a compliant stack sample from last week and a worker complaint about fumes at today's loadout. The supervisor proposes closing the complaint using the stack result.

Think it through.

The records differ in location, time and purpose. The stack result does not resolve today's worker exposure. Apply the workplace reporting/assessment process, control the affected task as required and retain both records with their identities. Investigate the actual loadout conditions and any process change without inventing a concentration.

Try it and explain your thinking

Trace dirty gas, filtered gas and collected dust through the baghouse drawing

Trace dirty gas, filtered gas and collected dust through the baghouse drawing. Explain one pollutant-control claim the drawing cannot support.

Skills you use: Reading, Communication.

Create a jurisdiction-and-record table for a fictional plant inside Metro Vancouver and one outside

Create a jurisdiction-and-record table for a fictional plant inside Metro Vancouver and one outside. Keep environmental and worker-exposure records in separate columns.

Skills you use: Writing, Problem solving.

Save a plume/complaint observation form

Save a plume/complaint observation form. A partner adds a change of fuel and a different time interval; revise the questions and explain the missing evidence.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use actual site authorization and exposure programs under qualified supervision.	0 / 1 / 2	Evidence/help: _____
Assess correct record/jurisdiction selection and explanation of dust-control components.	0 / 1 / 2	Evidence/help: _____
Emission testing, permit interpretation and occupational exposure assessment require the applicable competent professionals.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, D3, printed p. 43; source PDF pp. 45–45.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 7: Manufacture of Hot Mix Asphalt; source PDF pp. 341–390. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Asphalt Plant Regulation, B.C. Reg. 217/97. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 5: Chemical and Biological Substances. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 6: Substance Specific Requirements. Open publication.

- Metro Vancouver. (n.d.). Air Quality Regulatory Program. Current program locator, accessed 30 October 2026; verify actual permit requirements. Open publication.

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Study chapter · Outline E1

Determine asphalt mix design

A **mix design** establishes a suitable combination of aggregate, binder and other permitted ingredients for the required pavement performance. The accepted **job mix formula (JMF)** then provides the production target under the contract. It is more than a recipe by eye: the materials, proportions, test basis and performance requirements belong together.

- Asphalt/aggregate composition
- Types of mix designs
- Mix temperature characteristics

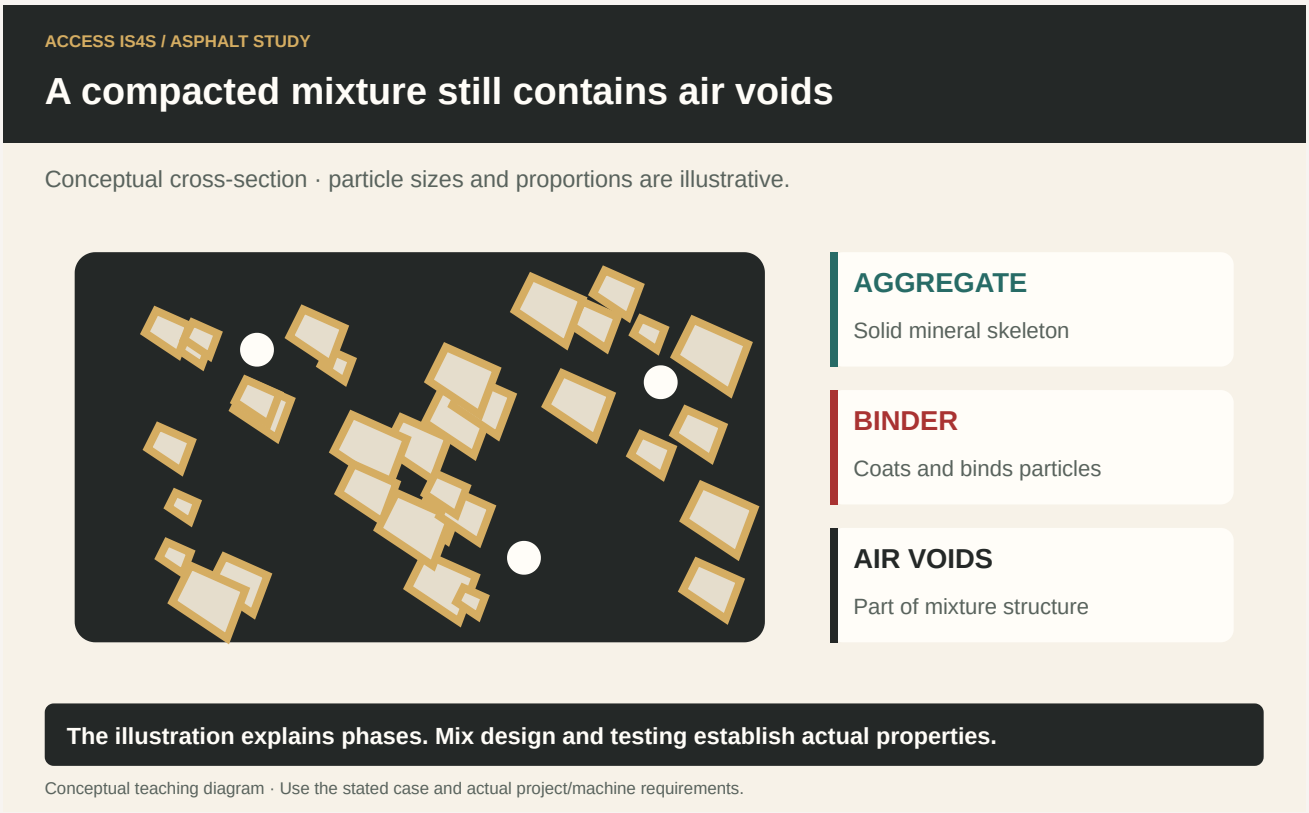
LEARNING STEP 1

Asphalt/aggregate composition

The aggregate skeleton, binder and air voids form a mixture whose response depends on both materials and compaction. **Marshall** design uses its specified specimen preparation and stability/flow and volumetric evaluations. **Superpave** design uses its specified material selection, gyratory compaction and volumetric framework, with additional tests where required. Their results and specimen methods are not interchangeable labels.

A field crew does not redesign a mixture by adding binder when the surface looks dry. Document the observation, identify the supplied JMF and refer the condition through quality control. Changes in aggregate source, binder grade or recovered material can affect the design and may require the specified approval process.

A compacted mixture still contains air voids



Note. ACCESS IS4S. (2026). A compacted mixture still contains air voids [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Types of mix designs

Dense-graded, open-graded, gap-graded and other designed mixtures distribute aggregate sizes differently and may serve different functions. A mixture chosen for drainage is not automatically suitable for every structural or traffic demand. The specification identifies the required mixture type, nominal size, binder and other design properties; the accepted formula identifies the actual approved combination.

Read the formula's product/source identities, target proportions, binder grade/content, recovered material where permitted, test basis and version. A production ticket with the wrong JMF identifier is a traceability issue even if the mixture looks familiar. Keep design approval separate from daily production conformity.

LEARNING STEP 3

Mix temperature characteristics

Hot-mix asphalt is produced and placed under its required elevated-temperature process. Warm-mix technologies permit specified production/placement approaches at reduced temperatures relative to the corresponding conventional process. Cold-mix products use a different material/process arrangement and are selected for the intended application. These categories do not supply one universal upper/lower temperature.

The approved mix, binder, equipment, haul, ambient conditions and placement/compaction procedure determine the usable temperature requirements. Read actual values from the governing documents and measure by the specified method. A surface infrared reading and an internal probe reading answer different questions. Temperature is recorded with location, time, method and instrument identity.

Marshall compaction hammer

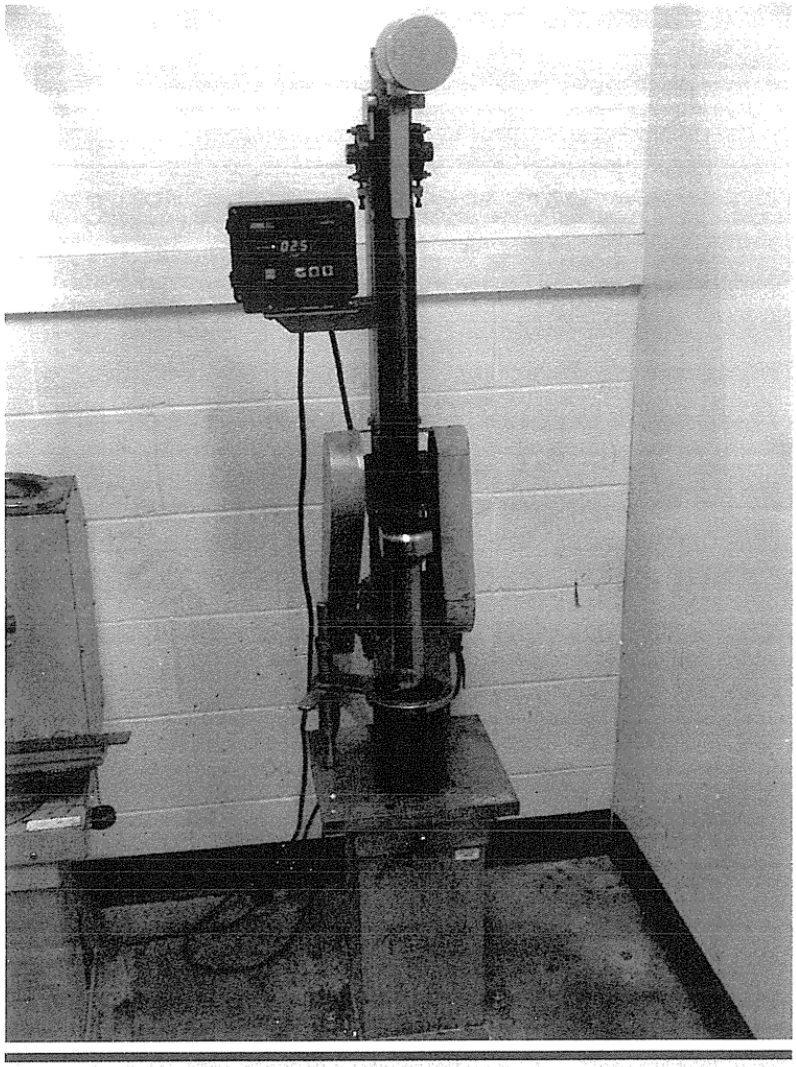


Figure 4.18 Marshall Compaction Hammer

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 4.18, p. 199; PDF p. 233). Faithful scan crop; original figure caption retained. Full source context.

Worked example · Keep the percentage basis

Fictional teaching JMF uses 5.2% binder by total mixture mass. A 25.0 t batch is considered for arithmetic only. No production temperature or field design is prescribed.

Think it through.

Binder model mass = 25.0 × .052 = 1.30 t; aggregate/other stated remainder = 23.70 t if no other components are separately specified. If the percentage were instead based on aggregate mass, this calculation would be different. Confirm the stated basis before comparing records.

Try it and explain your thinking

Explain Marshall and Superpave as different specified design/testing frameworks

Explain Marshall and Superpave as different specified design/testing frameworks. List five JMF fields needed by a production/placement crew.

Skills you use: Reading, Communication.

Recalculate binder model mass for 32.0 t at 5.2%

Recalculate binder model mass for 32.0 t at 5.2%. Explain why changing the mass percentage in the field is not an authorized design change.

Skills you use: Numeracy, Writing, Problem solving.

Build a digital JMF-reading card and have a partner swap the version identifier

Build a digital JMF-reading card and have a partner swap the version identifier. Revise the delivery check and record what must be confirmed.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use an authorized real JMF or the complete fictional arithmetic card.	0 / 1 / 2	Evidence/help: _____
Assess accurate identification of mixture, design method, percentage basis and temperature measurement requirements.	0 / 1 / 2	Evidence/help: _____
Mix design approval and changes are assessed by the qualified design/quality process.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E1, printed p. 44; source PDF pp. 46–46.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 4: Hot Mix Asphalt (HMA) Design; source PDF pp. 193–280. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.

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Study chapter · Outline E2

Confirm that samples match mix design

A test result belongs to a sample, and a sample must represent an identified material and interval. **Sampling plan**, **increment**, **composite sample**, **sample reduction**, **container**, **chain of custody** and **test method** are connected terms. A precise laboratory number is not useful if the wrong material was sampled or its identity was lost.

- Aggregate sampling
- Belt sampling
- Lab tests
- The process of taking mix temperature readings

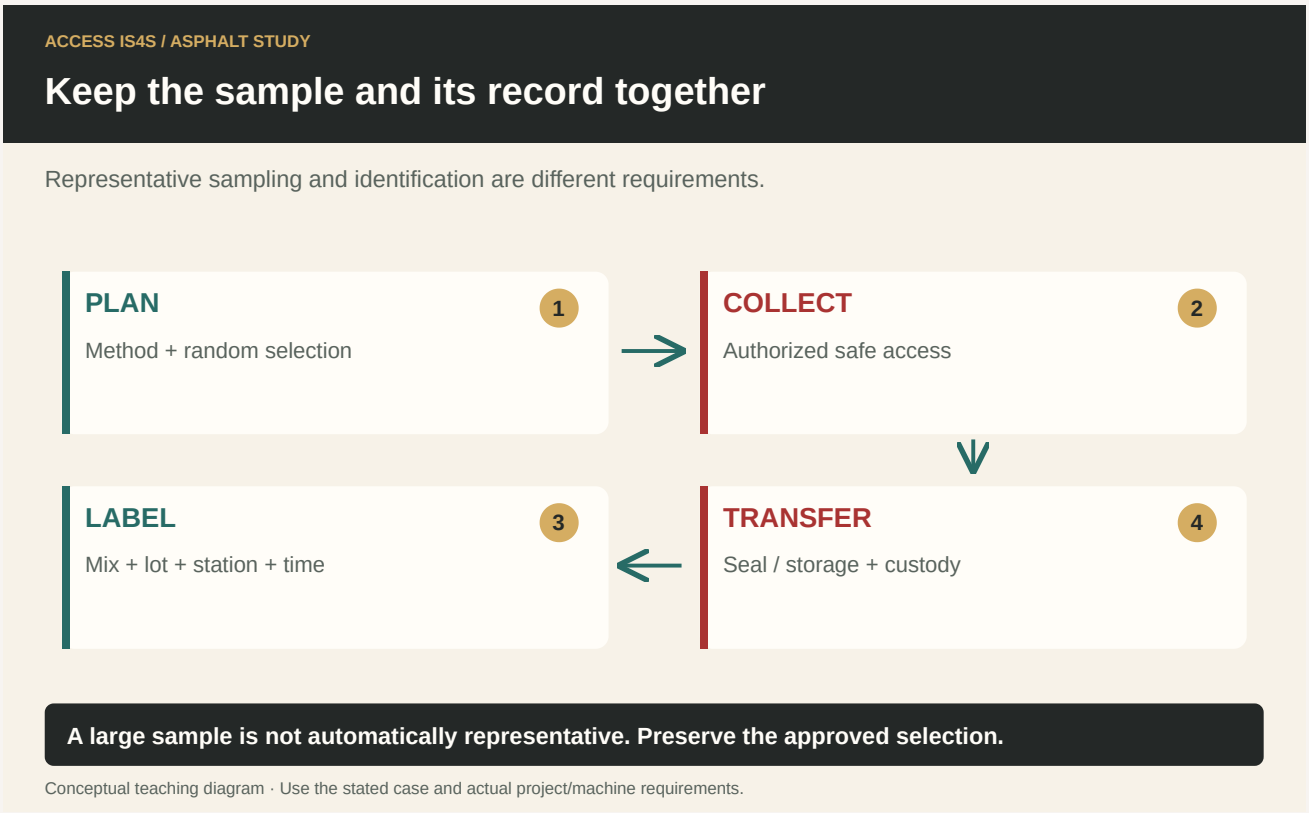
LEARNING STEP 1

Aggregate sampling

Aggregate sampling must follow the specified location, timing, number of increments and method. A convenient scoop from the visible top of a pile may not represent the material being supplied. Weigh-bin or other plant sampling requires the designed safe access and procedure; exposure to hot material, gravity and connected equipment must be controlled.

Preserve product/source identity and the relevant production interval. Combine or reduce samples only as the method permits. Record unusual conditions and contamination. A sample is not made representative by increasing its label detail after an unsuitable collection.

Keep the sample and its record together



Note. ACCESS IS4S. (2026). Keep the sample and its record together [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Belt sampling

Belt sampling can characterize a material stream when performed under the approved method and safe arrangement. A worker must not reach across or into a live moving belt. The required isolation/access and sampling device/procedure determine how the sample is obtained. Include the full required stream/fraction rather than selecting the pieces that look most typical.

A drum-plant process record connects the sampled feed with its bin, proportioning and time. A stopped-belt method and an approved automatic sampling device have different controls; neither is replaced by a generic drawing. Keep the sample method and the safe-work method together.

LEARNING STEP 3

Lab tests

Laboratory tests can determine gradation, binder content and mixture volumetric or other specified properties. The result must identify the method, units, sample and applicable comparison. Method differences and corrections can affect interpretation. A result outside a limit prompts the defined quality process; it does not justify editing the result or replacing it with a more convenient sample.

Distinguish a JMF target, production variability and the actual contractual acceptance criteria. Quality-control results help manage the process, while acceptance/audit results serve their specified role. Retain original results, corrections and disposition records.

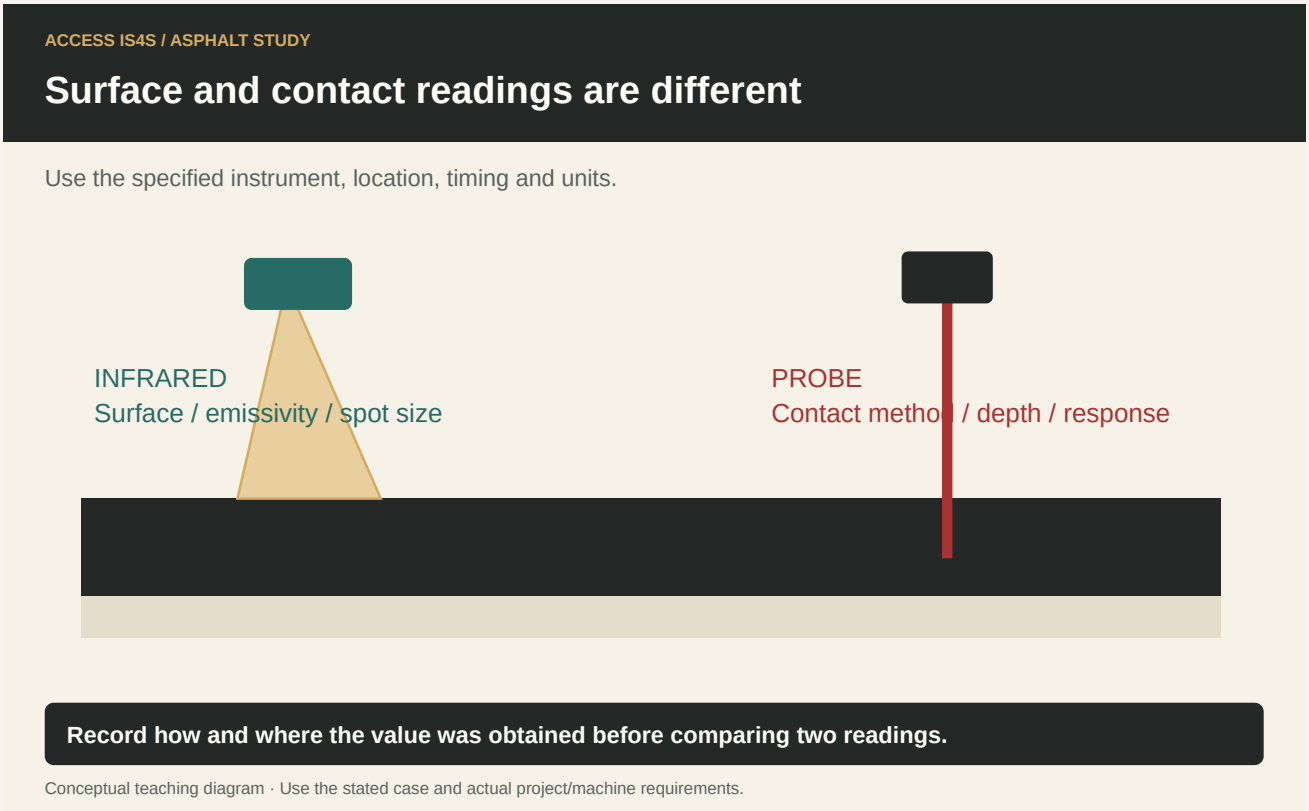
LEARNING STEP 4

The process of taking mix temperature readings

Select the temperature instrument and method for the required measurement. An infrared device senses a surface within its field of view; spot size, viewing conditions and surface properties affect the reading. A probe measures at its specified contact/insertion location. Neither automatically represents the entire load.

Verify instrument identity, condition and required check/calibration status. Record the time, location, product/load, units and method. If readings disagree, investigate the different measurement conditions before averaging them. A Celsius/Fahrenheit conversion changes units, not where the temperature was measured.

Surface and contact readings are different



Note. ACCESS IS4S. (2026). Surface and contact readings are different [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A result with a broken chain

Fictional bag label says aggregate bin A, 09:15. The sampling log says bin B, 09:15, and the test sheet uses the A label. The test values are within the limits printed on the teaching card.

Think it through.

The identity conflict prevents a reliable match to the intended material. Preserve all records, hold the interpretation and resolve the sampling chain through the responsible quality role. Passing numbers do not repair identity. If the source cannot be established, the required resampling/disposition process must be used.

Try it and explain your thinking

Build a sample route from planned location to test report

Build a sample route from planned location to test report. Identify where representativeness, safety and identity can each fail.

Skills you use: Reading, Problem solving, Communication.

Convert a fictional recorded 140°C to Fahrenheit using F = 1.8C + 32

Convert a fictional recorded 140°C to Fahrenheit using F = 1.8C + 32. Explain why a 140°C surface reading is not proof of the internal temperature.

Skills you use: Numeracy, Writing.

Save a sample register for the worked case

Save a sample register for the worked case. A partner supplies a corrected log with an author/date; retain the original and document the resolution rather than overwriting history.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use the actual approved sampling and hazardous-energy/access procedures with qualified supervision.	0 / 1 / 2	Evidence/help: _____
Observe representative collection/reduction, contamination prevention, labelling and temperature method/recording.	0 / 1 / 2	Evidence/help: _____
Do not stage live-belt access or hot-material handling without the complete authorized arrangement.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E2, printed p. 45; source PDF pp. 47–47.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline E3

Determine haul truck samples

A truck sample connects the delivered mixture to a production and quality record. The truck, load ticket, mix identifier, time and sampled location must remain traceable. Sampling should answer the planned question; choosing only the easiest or coolest-looking portion can introduce bias. Safe access is established before sampling, not improvised after a truck arrives.

- Mix sampling
- Quality control
- Bag samples

LEARNING STEP 1

Mix sampling

Confirm the intended material and sampling purpose: mixture identity, temperature, binder-content/gradation testing or another specified property. Use the required sampling location, increments, tools and sample size under the approved method. The arrangement must control vehicle movement, hot material, access and other exposed people. Do not climb a load or reach into a discharge stream on the strength of a classroom description.

Temperature and laboratory specimens may require different handling. The sampling record identifies exactly what was taken, where, when and by whom. A result is compared with the correct formula and contractual criterion, not simply with the last truck’s appearance.

A selected load keeps its identity



Note. ACCESS IS4S. (2026). A selected load keeps its identity [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Quality control

Random selection reduces the opportunity to choose favourable loads or locations, while the approved plan determines the sampling population and timing. Random does not mean unplanned or unsafe. The testing agency/company personnel must have the required competence and authorization, and the truck/paver crew must understand the hold and communication arrangement.

Keep quality-control sampling separate from owner acceptance/audit sampling where the contract distinguishes them. A failed result is handled through the documented process, preserving the selected sample and records. Replacing it with an unrecorded convenient sample defeats traceability and can hide a production problem.

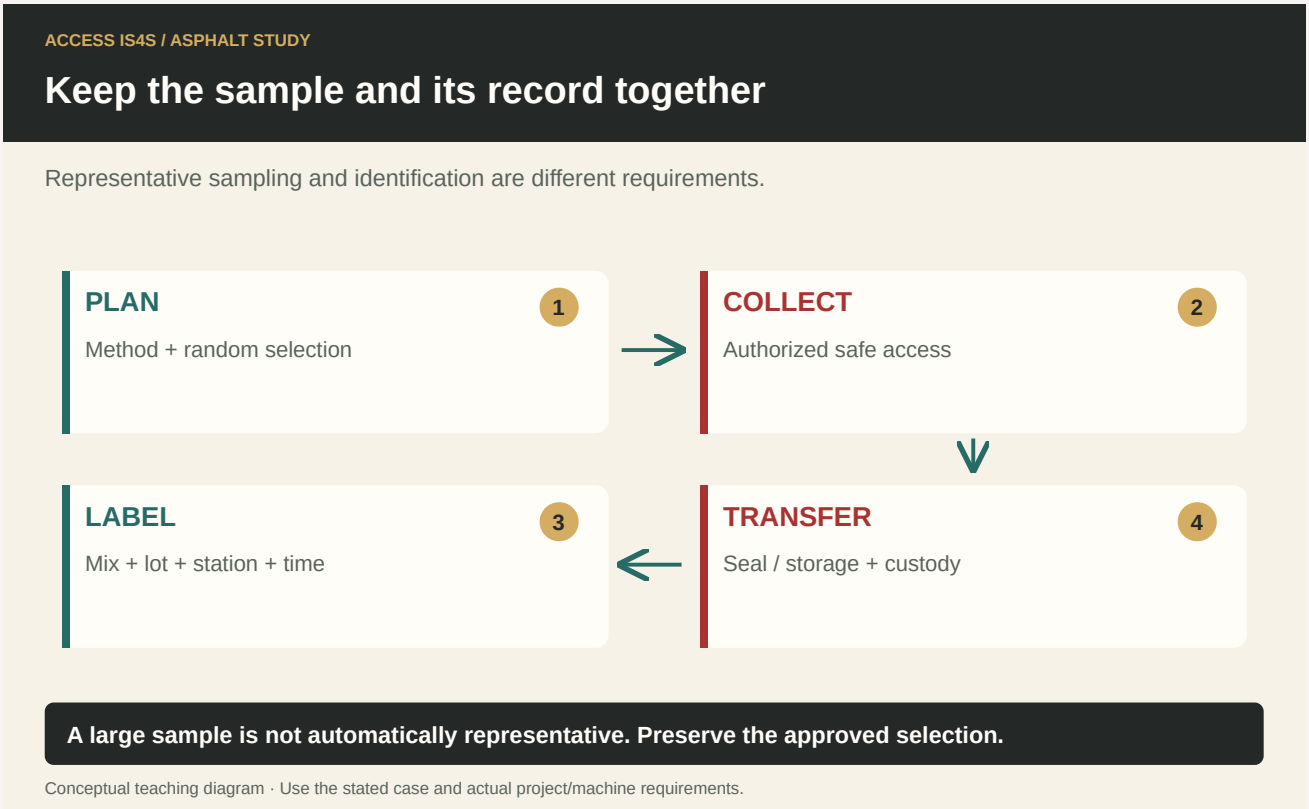
LEARNING STEP 3

Bag samples

Use the container or bag specified for the sample's material, temperature, quantity and test. Identify the load, mix, date/time, location, sampler, intended test and any required seal/backup identification. Protect the sample from contamination and loss while following safe hot-material handling. An ordinary classroom bag may be unsuitable for actual hot mix.

A backup sample is useful only if it remains correctly identified, preserved and handled under the applicable method and retention arrangement. Record transfer and receipt. If a seal breaks or the label becomes unreadable, report the condition rather than recreating identity from memory.

Keep the sample and its record together



Note. ACCESS IS4S. (2026). Keep the sample and its record together [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Random selection with traceability

Fictional teaching plan identifies loads 1–10 and a recorded random selection of loads 3 and 8. Load 8 is delayed; load 7 is easier to reach. The sampler proposes silently substituting 7.

Think it through.

The approved selection has changed. Follow the plan’s authorized contingency/rescheduling method and record any change; do not silently substitute a convenient load. Maintain safe coordination while waiting or arranging the approved alternative. Random selection and safe access must both be satisfied.

Try it and explain your thinking

Explain and apply the idea

Prepare the complete label and transfer record for fictional load 3, mix JMF-C revision 2, sampled at 10:25 for binder/gradation testing. Mark any information the case does not supply.

Skills you use: Reading, Writing.

Two fictional increments are 6.2 kg and 5.8 kg

Two fictional increments are 6.2 kg and 5.8 kg. Find total mass and explain why this sum alone cannot establish a representative or correctly reduced laboratory sample.

Skills you use: Numeracy, Problem solving, Communication.

Communicate and verify the work

Create a digital sampling log, introduce the delayed-load change with a partner and preserve the authorized decision trail.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Instructor establishes actual vehicle separation, access, tools, containers and sampling method with the competent testing role.	0 / 1 / 2	Evidence/help: _____
Observe representative sampling from authorized locations, identification, safe handling and backup/transfer procedure.	0 / 1 / 2	Evidence/help: _____
Record practical evidence from the actual task; paper labels alone do not demonstrate hot-mix sampling skill.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E3, printed p. 46; source PDF pp. 48–48.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Determine compaction levels

Density relates mass to volume. Compaction changes the arrangement of particles and air voids, but the measured density must be compared with the correct reference and method. Two instruments can report numbers that look similar while representing different locations or measurement principles. Keep the instrument, sample, units and denominator visible.

Two methods to determine compaction

LEARNING STEP 1

Two methods to determine compaction

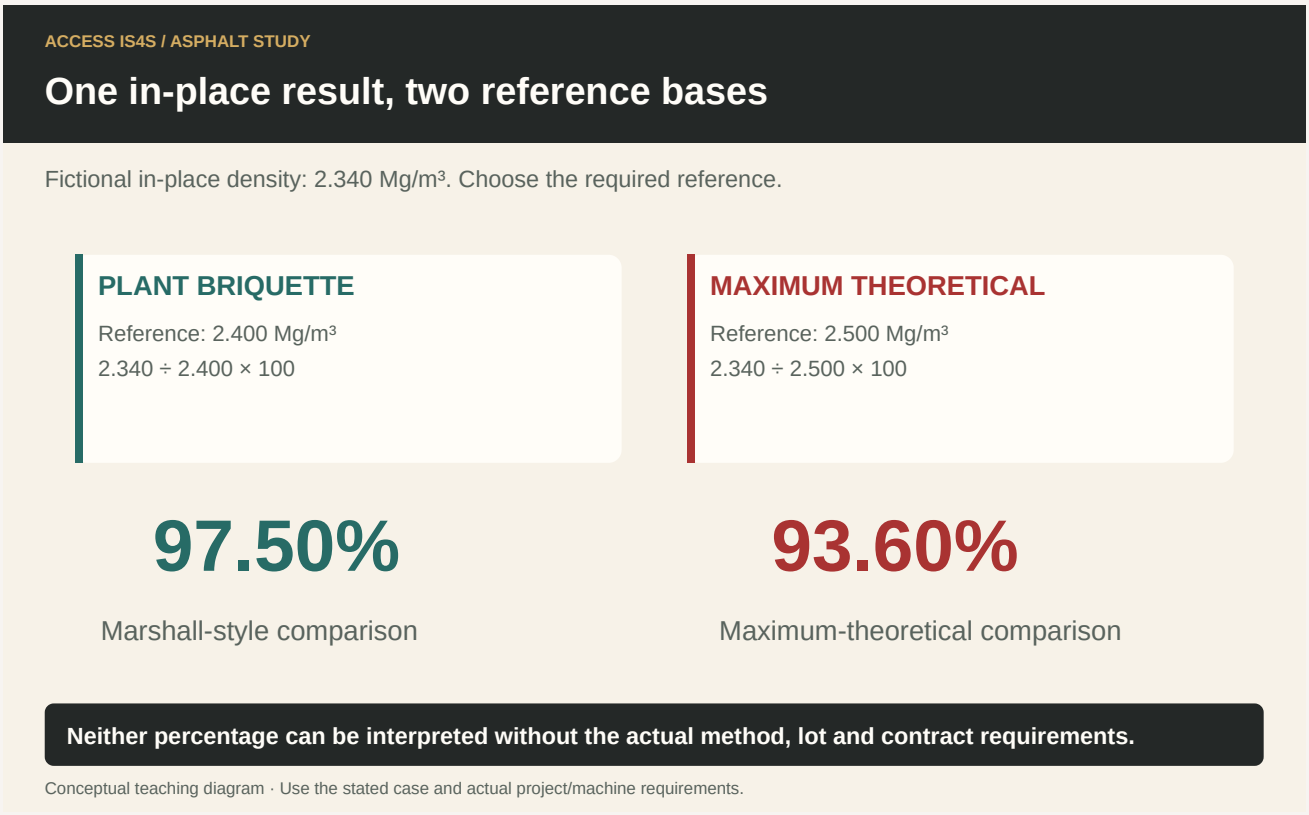
A **nuclear density gauge** uses a radioactive source and detector arrangement to infer a material property through its calibrated method. The actual instrument, mode, calibration/standardization, surface preparation and test method affect the result. In Canada, possession/use and transport require the applicable CNSC licensing, training, security and radiation-safety arrangements. Reading the Troxler 3430 manual is useful instrument study; it does not authorize source handling or operation.

A **core** is a physical specimen removed at an identified location. The approved coring and laboratory method establish sampling, handling, dimensions and density determination. Coring also requires utility/location checks, traffic/access controls and safe equipment/water management, followed by the specified repair of the hole. A damaged, contaminated or mislabelled core may not support the intended result.

Gauge and core results are compared only through the specified procedure and any required correlation. A percentage density must state its reference: maximum theoretical mixture density, laboratory compacted reference or another defined basis. Do not compare percentages that use different denominators. The result belongs to the sampled location and acceptance unit defined by the contract; one point does not automatically represent an entire project.

Report original readings, checks, location and any limitation. A low result prompts the approved investigation/disposition process. Additional rolling is not automatically appropriate after the material's workable window or when the cause is unknown.

One in-place result, two reference bases



Note. ACCESS IS4S. (2026). One in-place result, two reference bases [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Troxler 3430 gauge and accessories

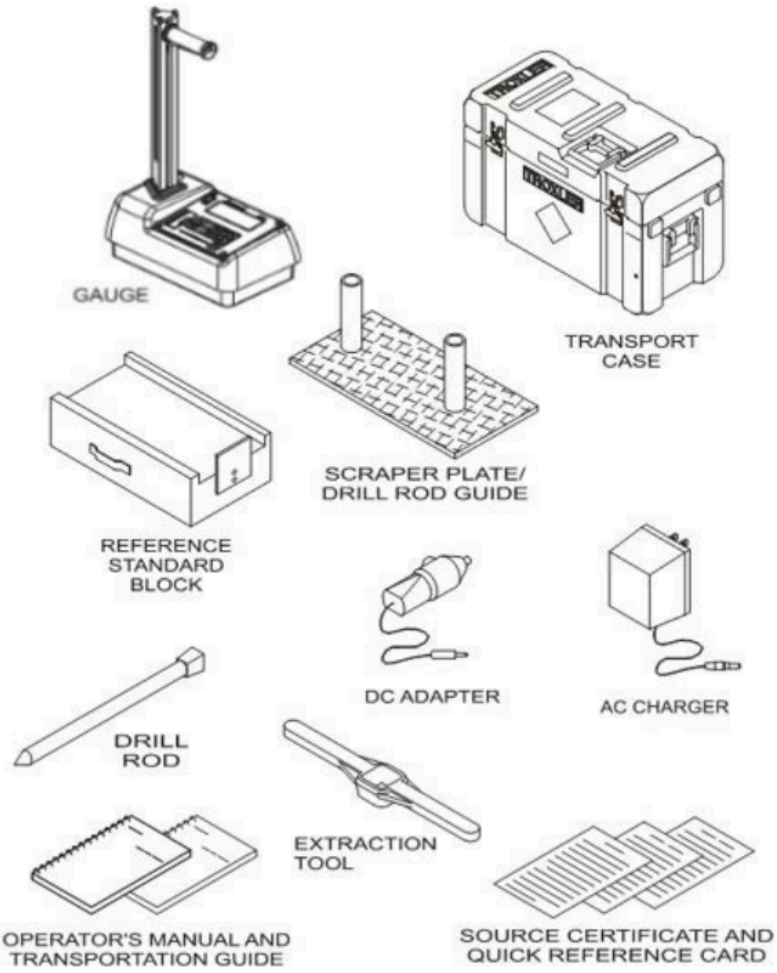


Figure 1. Model 3430 Gauge and Accessories

Note. Troxler Electronic Laboratories. (2021). RoadReader Model 3430: Manual of operation and instruction (Ed. 1.3, Fig. 1, p. 1-5; PDF p. 19). Identification only; Canadian use requires applicable CNSC authorization. Full source context.

Worked example · The denominator changes the meaning

Fictional in-place density is 2.34 t/m³. Reference A, maximum theoretical density, is 2.50 t/m³. Reference B, a compacted laboratory reference in a different teaching system, is 2.40 t/m³.

Think it through.

Against A: 2.34/2.50 × 100 = 93.6%. Against B: 2.34/2.40 × 100 = 97.5%. Both calculations use the same numerator; they answer different defined questions. Neither establishes acceptance without the applicable method and criterion. Never relabel 97.5% as percent of maximum theoretical density.

Try it and explain your thinking

Explain and apply the idea

Compare gauge and core methods using columns for measurement principle, identification, checks, access and authorization.

Skills you use: Reading, Writing, Problem solving.

Repeat the worked ratios for an in-place value of 2.30 t/m³

Repeat the worked ratios for an in-place value of 2.30 t/m³. Write each full label.

Skills you use: Numeracy, Communication.

Communicate and verify the work

Build a saved density-result sheet and peer-test whether its denominator, units and test location can be found immediately. Correct any ambiguity.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Check your understanding of density evidence

Five questions, four marks each: 20 marks total. Minimum 14/20 (70%), as required by the outline for this knowledge check. This does not authorize gauge operation.

Question 1 · Explain one difference between core testing and nuclear-gauge testing, and identify the Canadian authorization issue.

Question 2 · Calculate percent density for 2.30 Mg/m³ in place and 2.50 Mg/m³ maximum theoretical.

Question 3 · Name four traceability fields needed with a density result.

Question 4 · A draft changes the reference density to improve the percentage. Explain the response.

Question 5 · Plan a safe learning observation of core or gauge testing.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use a licensed/authorized gauge arrangement or approved core-sampling/laboratory arrangement with qualified supervision.	0 / 1 / 2	Evidence/help: _____
Assess method recognition, traceability, calculation and correct interpretation; no learner handles a radiation source without required authorization.	0 / 1 / 2	Evidence/help: _____

What the observer checks	Result	Evidence
For the outline knowledge quiz, use the questions in this unit with the provider's grading record; practical testing remains a separately observed outcome.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E4, printed p. 47; source PDF pp. 49–49.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- Troxler Electronic Laboratories. (Edition 1.3, February 2021). RoadReader Model 3430 Surface Moisture-Density Gauge: Manual of Operation and Instruction. Open publication.
- Canadian Nuclear Safety Commission. (Web snapshot: October 30, 2026). Portable Nuclear Gauges. Open publication.
- Canadian Nuclear Safety Commission. (n.d.; current publisher download). Portable Gauge Quick Reference Guide. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Integrated practice · all nine Skills for Success

Trace a material and its sample

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional 1,200 g aggregate sample has retained masses of 180 g, 420 g, 360 g and 240 g in the pan. Its log says Mix M, lot 4, but the container label says lot 5. A draft report says “the sample is representative because it is large” and “20% RAP means 20% lower emissions.” The project location is in Metro Vancouver. No acceptance limits or permit number are supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Change card · Use after the first attempt

The quality lead supplies a corrected source record linking the container to lot 5, with documented authorization and preserved original record.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person's actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.